Cold Extrusion

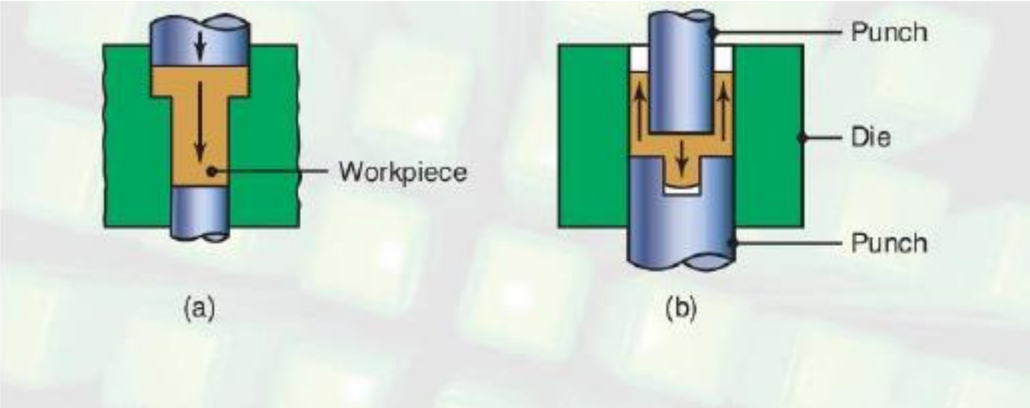
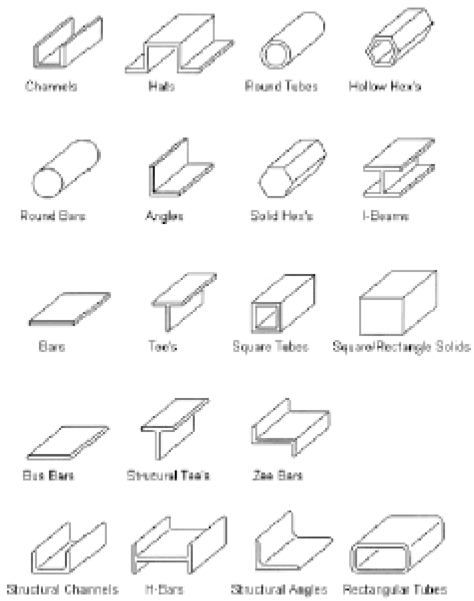


FIGURE 6.54 Two examples of cold extrusion. Arrows indicate the direction of material flow. These parts may also be considered as forgings.

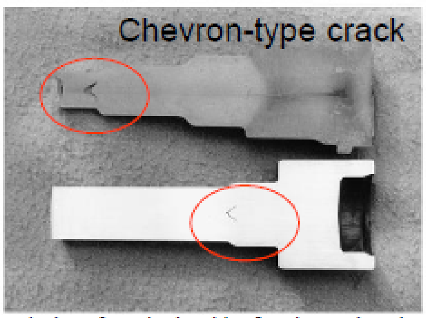
Extruded parts

- Long products, uniform section
- Section can be complex (difficult by rolling)
- Small billets are used for cans
- Ductile materials (aluminum)



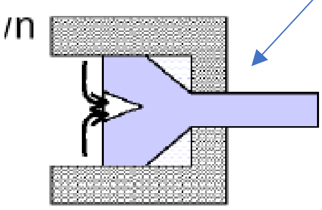
Extrusion defects

- Defects due to internal tensile stress state (similar to necking)
- Lubrication improves the problem
- Impurities and the extrusion angle make the problem worse

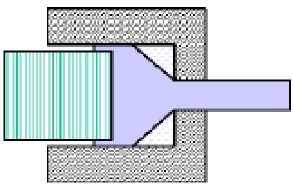


Extruded surface (solved by finish maching)

- Material surface
- Core shrinkage
- (crack, shrinkage)



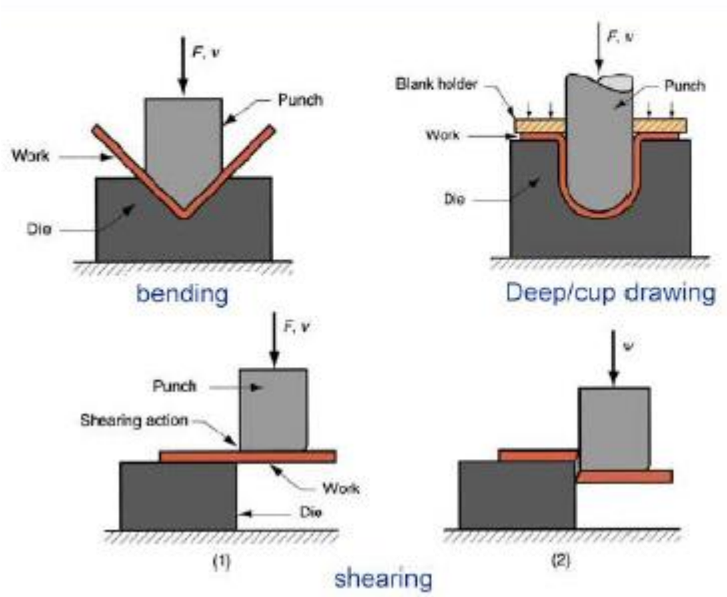
Flow of material not enough to fill the cavity



Extrusion workshop

Sheet forming

It is a very important process since most parts produced in the industry are sheets.

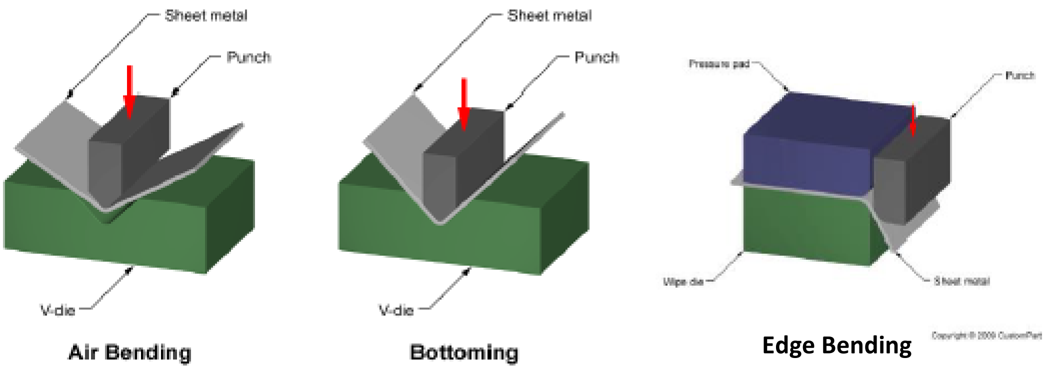


Bending → the punch of a specified angle is pressed against the sheet lying within a cavity of the same angle.

Deep/cup drawing → force is not too high, to allow the material to slide and take the shape.

Shearing → sheet cut without any bending or defects or distortion or stresses.

Methods of bending



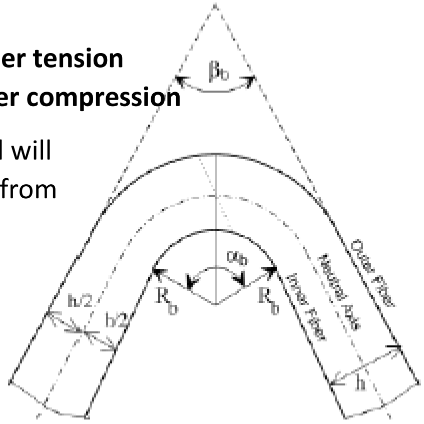
Air Bending → there is no constraint force on the sheets → sheet free to bend, to follow the falling of the punch.

Bottoming → sheet pressed against the surface of the die
Edge Bending → no control on the angle, slide responsible for bending.

<u>Bending: a crude means to test material sheet performance</u>

Outer fiber under tension
Inner fiber under compression

If the material will fail, it will fail from the outside



ℓ_{NA} = length of neutral axis (dashed line)
ℓ_{OF} = length of outer fiber

Bending angle

$$\alpha_b = \frac{\ell_{NA}}{R_b + \frac{h}{2}} = \frac{\ell_{OF}}{R_b + h}$$

thickness

Bending Strain:

$$\varepsilon_{OF}^{tensile} = \frac{\ell_{OF} - \ell_{NA}}{\ell_{NA}} = \frac{1}{\frac{2 \cdot R_b}{h} + 1}$$

Bending radius: the smallest radius that a metal sheet material can bend around

$R_{b,min} = h\,(50/\Delta - 1)$

Δ= %area of contraction after tensile test
(if Δ = 50%, $R_{b,min}$ =0, the sheet bends over itself)

Δ >> if + heating or
if plastic deformation assisted by pressure

Limit radius for which the material can be bent without cracking
With the tensile test, the material will shrink and contract because of pulling → contraction area

<u>Bending</u>
If R_b is ***generous***, the neutral line is in the ***center***.
If R_b is ***tight*** (severe), the neutral line shifts towards the ***compressive side***. In this case, the centerline is elongated and volume constancy is preserved by ***thinning***! The increased length of the centerline is taken into account for $R_b < 2h$ by assuming that the neutral line is located at one-third of the sheet thickness.
When the sheet is relatively narrow (w/h < 8), there is also a contraction in width w.

<u>Bending: limitations</u>
We can do a test on the material to access its formability but also limitations.

1) Orange Peel

2) Crushing on inside surface

3) Minimum Bend Radius:

To avoid thinning

For no necking: $$\varepsilon_{eng}^{tensile} = \frac{1}{\frac{2 \cdot R_b}{h} + 1} < \varepsilon_{eng,u} \approx e^n - 1 \longrightarrow \boxed{R_b > \frac{2 - e^n}{2(e^n - 1)} \cdot h}$$

For no fracture: $$\boxed{R_b > \left(\frac{1}{2q} - 1\right) \cdot h \quad \text{for } q < 0.2}$$ (less ductile materials)

$$\boxed{R_b > \left(\frac{(1-q)^2}{2q - q^2}\right) \cdot h \quad \text{for } q \geq 0.2}$$ (more ductile materials)

Orange Peel: defect associated with a rough surface appearance after forming a component from sheet metal. The surface has the appearance of the surface of an orange.
Larger R_b and h → the material is no good: bending is worse.

Bending radius for various alloys

The material is annealed

The material after it has been heat treated

Material	Condition Soft	Condition Hard
Aluminum alloys	0	6T
Beryllium copper	0	4T
Brass, low-leaded	0	2T
Magnesium	5T	13T
Steels		
Austenitic stainless	0.5T	6T
Low-carbon, low-alloy, and HSLA	0.5T	4T
Titanium	0.7T	3T
Titanium alloys	2.6T	4T

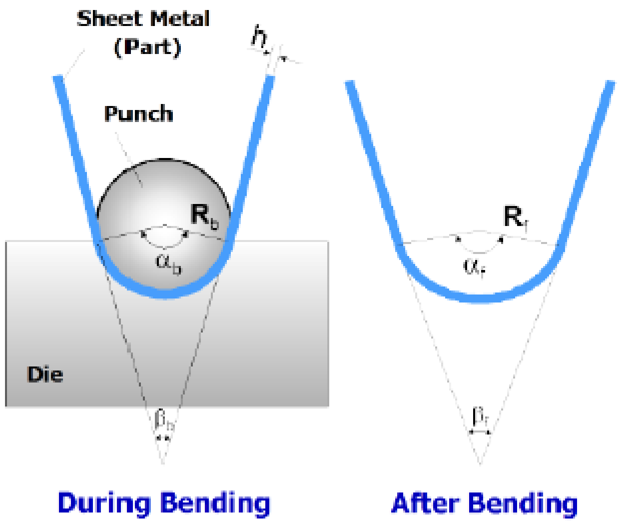
High content of Nickel

Higher resistance

T = thickness

Beryllium copper is more ductile and formable than Al.
The c lattice constant measures separation between atoms in the basal planes; the ideal hexagonal ratio is 1,63.
Magnesium is brittle since it has hexagonal crystal structure.
Magnesium: the c lattice constant is larger than ideal; the two basal planes are more separated from each other → when polycrystals are sheared by plastic deformation, the planes will slide and break the material.
Titanium is even better than steels since it has hexagonal crystal structure (like Mg).
Titanium: ration less than ideal. The two basal planes are constrained over the middle plane over the hexagon → atoms are pressed to each other; when there's sliding, planes shear without breaking the bonding.
The high separation between basal planes leads to weakness of the crystal.
Hexagonal structure → cannot absorb energy → brittle.

Bending: springback



R_b: radius of bending die
α_b: bending angle
β_b: included die angle

R_f: radius of bent part
α_f: final bend angle
β_f: final included part angle

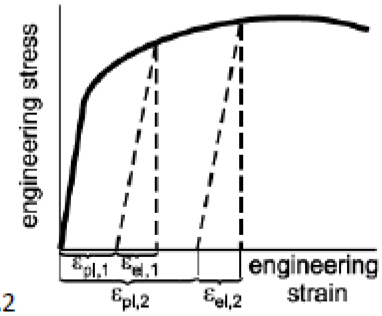
Note: $\alpha_b + \beta_b = 180°$
$\alpha_f + \beta_f = 180°$

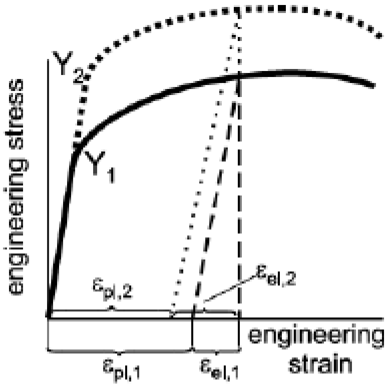
The sheet relaxes a little bit after the punch is release → recovery of the shape → **springback**: the amount of elastic return of the shape of the plastically deformed sheet. This is one of the worst problems in automotive

Springback from tensile curve

- Factor n.1

$$\frac{R_b}{h}\uparrow \;\Rightarrow\; \frac{R_b}{R_f}\downarrow \;\therefore\; \text{springback}\uparrow$$





Factor n.2

$$Y\uparrow \;\Rightarrow\; \frac{R_b}{R_f}\downarrow \;\therefore\; \text{springback}\uparrow$$

Factor n.3

$$E\uparrow \;\Rightarrow\; \frac{R_b}{R_f}\uparrow \;\therefore\; \text{springback}\downarrow$$

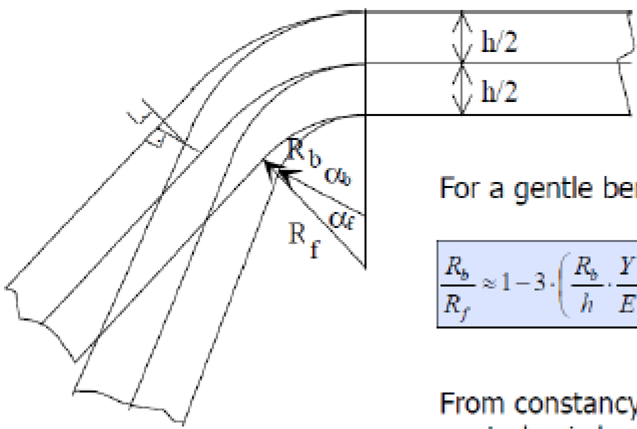
In case we use different material, not steel

The two curves have the same elastic modulus (same slope), since they are both steels, but they have different σ_y (Y) and different UTS.

If we load both steels up to UTS and then unload, they will follow the unloading lines, which are parallel (they have the same slope). The springback can be read on the horizontal axis.

The larger σ_y and UTS, the larger the springback.

Springback in bending



For a gentle bend and a high Y/E:

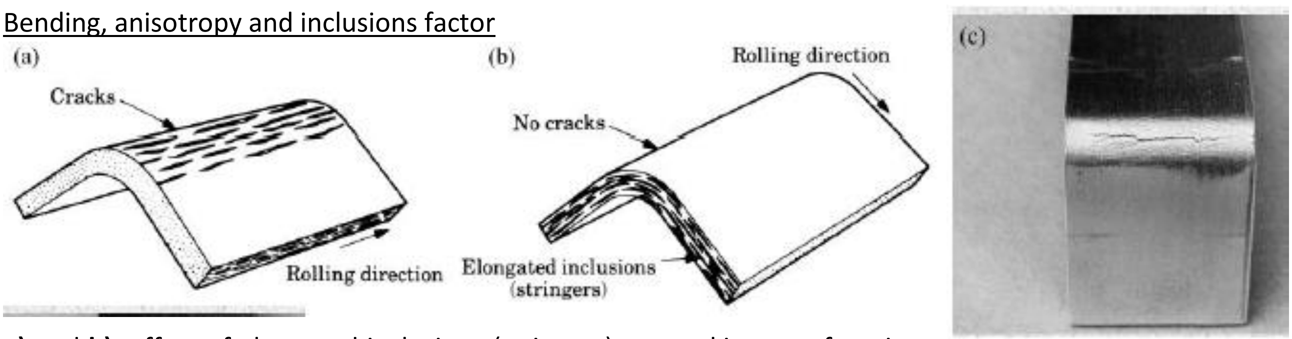
$$\frac{R_b}{R_f} \approx 1 - 3\cdot\left(\frac{R_b}{h}\cdot\frac{Y}{E}\right) + 4\cdot\left(\frac{R_b}{h}\cdot\frac{Y}{E}\right)^3 \quad < 1 \text{ (always)}$$

From constancy of neutral axis length:

$$\alpha_f = \left(\frac{R_b + \frac{h}{2}}{R_f + \frac{h}{2}}\right)\cdot\alpha_b$$

$R_f < R_b$ → springback observed

Bending, anisotropy and inclusions factor



a) and **b)**: effect of elongated inclusions (stringers) on cracking as a function of the direction of bending with respect to the original rolling direction of the sheet

c) Cracks on the outer surface of the strip bent to an angle of 90°. Note the narrowing of the top surface due to Poisson's effects.
We bend the material in the less favorable direction if we want to see if the material is too weak in bending and in other formability processes. If we have crack, the material is unsuitable for sheet forming, because the material will strain in this condition and break.

Drawing vs. stretching

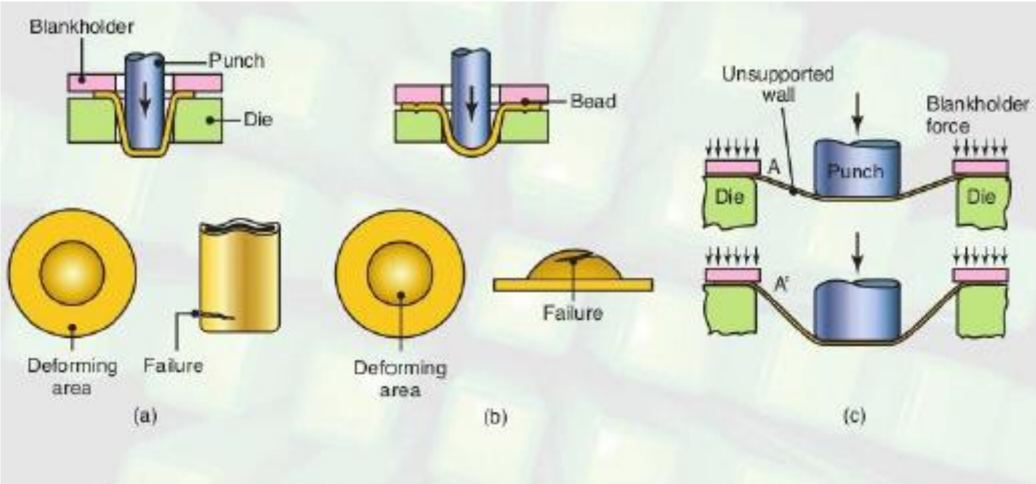
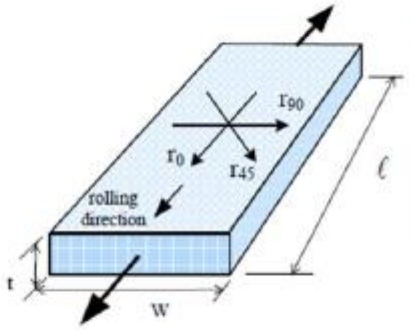


FIGURE 7.51 Examples of (a) pure drawing and (b) pure stretching; the bead prevents the sheet metal from flowing freely into the die cavity. (c) Unsupported wall and possibility of wrinkling of a sheet in drawing. Source: After W.F. Hosford and R.M. Caddell.

During drawing, the metal is flowing in the die and the blank is changing the shape. During stretching, there is not an inward movement of the blank edge. The surface area of the blank is displaced by tension in the different shape via control metal flow.
Stretching: the surface area of the blank is increased as a result of tension and thinning of the metal. The blank holder doesn't allow any sliding of the sheet.
Stretching is also called **thinning** because any stretching will make the material thinner. We never have pure drawing or pure stretching in deep drawing in real life, but we have a combination of both

Anisotropy: definitions



Volume constancy requires:

$$t_0 \cdot w_0 \cdot \ell_0 = t_1 \cdot w_1 \cdot \ell_1 \quad \text{or} \quad \frac{t_1}{t_0} \cdot \frac{w_1}{w_0} \cdot \frac{\ell_1}{\ell_0} = 1$$

Taking the logarithm:

$$\ln\left(\frac{t_1}{t_0} \cdot \frac{w_1}{w_0} \cdot \frac{\ell_1}{\ell_0}\right) = \ln(1) = 0$$

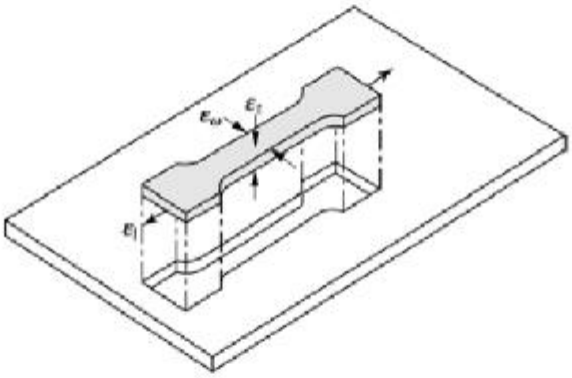
yields $\varepsilon_t + \varepsilon_w + \varepsilon_\ell = 0$ where:

$\varepsilon_t = \ln\left(\frac{t_1}{t_0}\right)$ (thickness strain) $\quad \varepsilon_w = \ln\left(\frac{w_1}{w_0}\right)$ (width strain) $\quad \varepsilon_\ell = \ln\left(\frac{\ell_1}{\ell_0}\right)$ (length strain)

Normal Anisotropy

resistance to thinning

Strains (ε) in a tensile test specimen taken from blank hot-or-cold rolled sheet:



Crystallographic anisotropy: elongated grains.

Mechanical anisotropy: inclusion fibers along the thickness.

Normal anisotropy: $R = \varepsilon_w / \varepsilon_t$

Reminder: $\varepsilon_l + \varepsilon_w + \varepsilon_t = 0$

For pure tensile loading: $R = 1$

Normal anisotropy R determines the sheet thinning during sheet forming; very important in deep drawing.

R can be measured by tensile test.

The two strains ε_w and ε_l aid estimate the normal and planar anisotropy of the material.

In principal directions, we have no shear. The larger the R, the better the formability because there is no thinning.

When ε_t is equal to 0, R goes to infinity.

Average normal anisotropy $\bar{R}$

•The heterogeneous thinning of a metallic sheet during forming is controlled by the average normal anisotropy $\bar{R}$

• Anisotropy means that properties are direction dependent; in the case of rolled parts, this directional dependence is imparted by the rolling direction

• The rolling direction makes the properties significantly different between the rolling and its transversal (90°) direction

• $\bar{R}$ is defined as a function of true strains that can be measured by a standard tensile test; the specimen is taken along the rolling (0°), the transversal (90°) and the intermediate (45°) direction

• Typically, to determine $\bar{R}$ elongation as large as 15-20% are considered by convention

• In order a material to be drawn $\bar{R}$ has to be at least 1

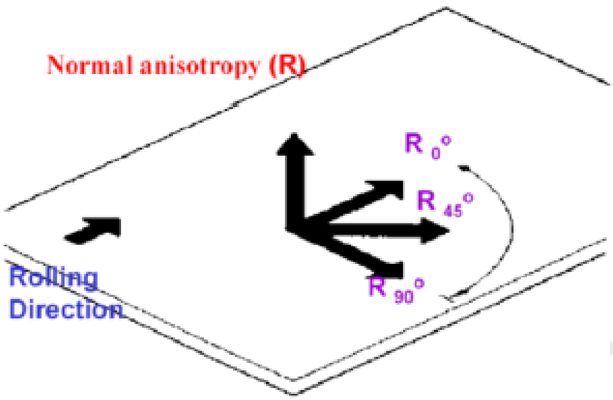
• which means that the strain along the thickness has to be either comparable to or smaller than the strain along the width; that is, the thickness t should not reduce remarkably compared to w; this condition imposes that the material has to deform uniformly along the thickness in order to exploit effectively the sheet material

• An excessive reduction of the thickness compared to w will warn an undesirable susceptibility of the material to heterogeneous thinning

• For typical cold DEEP DRAWN sheets $\bar{R}$ ranges from 1 to 2

• $\bar{R}$ follows a linear relationship with LDR measured from deep drawing tests

• It ensures that R can be used as an effective sheet property for sheet forming and hence can be used as an effective formability property of the material



Isotropic material: $R = 1$ for each direction

Anisotropic material: $R \neq 1$

Normal anisotropy: $R_0 = R_{45} = R_{90} \neq 1$

Planar anisotropy: $R_0 \neq R_{45} \neq R_{90}$

Normal average anisotropy:

$$\bar{R} = R_{avg} = (R_0 + 2R_{45} + R_{90})/4$$

Planar average anisotropy:

$$\overline{\Delta R} = (R_0 - 2R_{45} + R_{90})/2$$

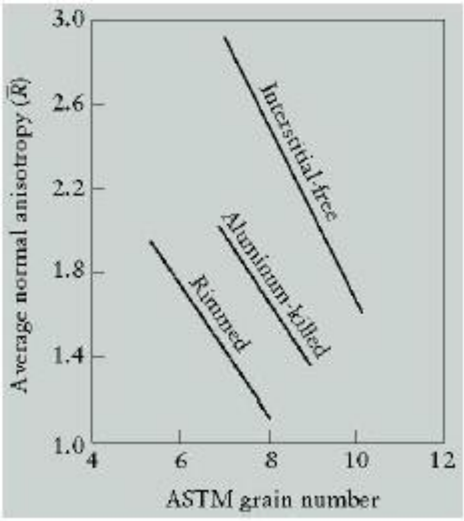
For good formability, $\bar{R} \geq 0.8$ (BCC) or $0.4 \leq \bar{R} \leq 0.8$ (FCC) and $\overline{\Delta R} = 0$

Normal anisotropy $\overline{R}$ for various sheet alloys

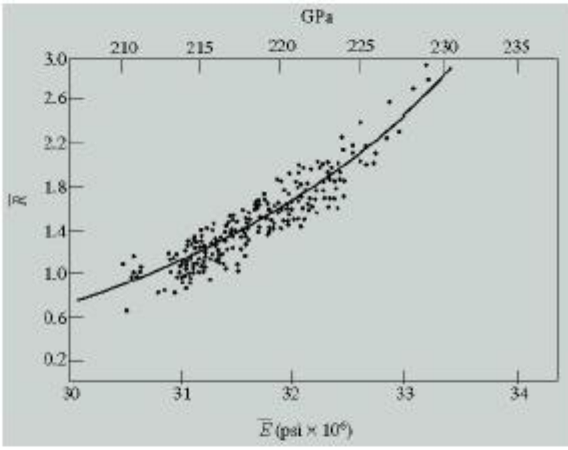
resistance to thinning	$\overline{R}$
Zinc alloys	0.4–0.6
Hot-rolled steel	0.8–1.0
Cold-rolled rimmed steel	1.0–1.4
Cold-rolled aluminum-killed steel	1.4–1.8
Aluminum alloys	0.6–0.8
Copper and brass	0.6–0.9
Titanium alloys (α)	3.0–5.0
Stainless steels	0.9–1.2
High-strength low-alloy steels	0.9–1.2

A material with low R cannot be shaped by drawing. Aluminum is very likely to thin and to neck during sheet forming, leading to failure → not very used since it's easy to fail in necking.
Titanium has a very high R → highly formable and highly ductile.

Normal average anisotropy $\overline{R}$



Effect of grain size on R for various low %C steels



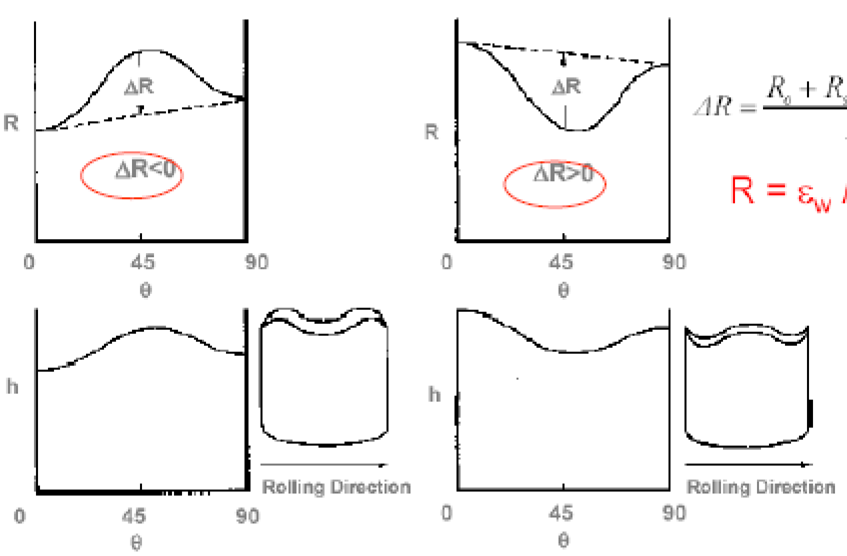
Relationship between R and Young modulus E, for steel sheets

Larger grain size is very good for formability (high R). Long-range displacement before reaching GB → the grain can be strained.
The larger the grain number, the lower grain size. Larger E leads to higher R.
Steel with no interstitials like C, N, may strain very highly (since R is high).
Low carbon steel contains an amount of iron oxide such that continuous generation of carbon dioxide during solidification is not inhibited, leading to porosity.
[**Killed steel** is a steel that had been completely deoxidized by the addition of an agent before casting so that there is practically no evolution of gas during solidification → high degree of chemical homogeneity and freedom from gas porosity. Common deoxidizing agents include aluminum, ferrosilicon, manganese, which are very reactive with O_2 → instead of O_2 we will have oxides. Formability is reduced by oxides because particles will tend to lock the grain, so the grains become smaller, which are not suitable for formability. Material strengthened by grain refinement.]

Induced defects by Planar anisotropy
•A non-zero planar anisotropy causes so-called earings at the upper border of the drawn cup
• If ΔR = 0, earings do not originate
•The earings' height increases with increasing ΔR
•The number of earings may be: 2, 4, 8

• For defect-free drawn cups: R >1; ΔR→0
• Alloying elements and rolling parameters
may influence these values.

Combined effect of $\overline{R}$ and ΔR



$$\Delta R = \frac{R_0 + R_{90} - 2R_{45}}{2}$$

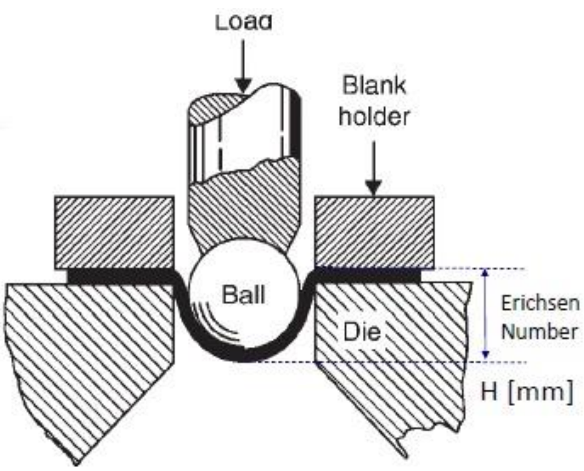
$$R = \varepsilon_w / \varepsilon_t$$

Bending radius vs. Alloy type

Material	Condition: Soft	Condition: Hard
Aluminum alloys	0	6T
Beryllium copper	0	4T
Brass, low-leaded	0	2T
Magnesium	5T	13T
Steels		
Austenitic stainless	0.5T	6T
Low-carbon, low-alloy, and HSLA	0.5T	4T
Titanium	0.7T	3T
Titanium alloys	2.6T	4T

Sheet forming test: Erichsen cupping Test, the oldest

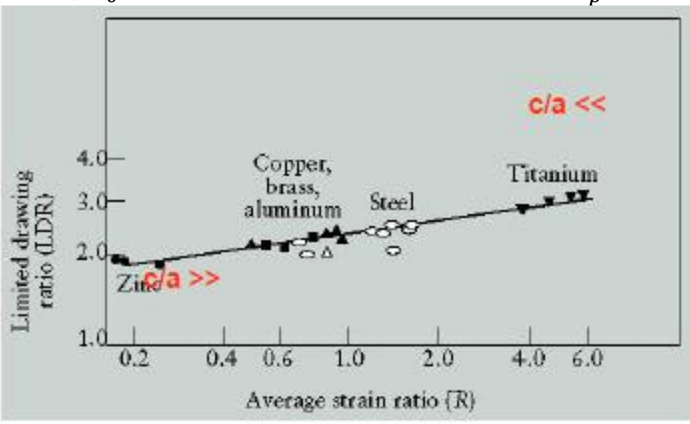
The test **gives some idea of the grain size** of the material, and hence its **suitability for deep-drawing**.
Coarse grain, always associated with **poor ductility in a drawing operation**, will show up as a **rough rumpled surface** in the dome of the test-piece.
This resembles the 'lumpy' outside skin of an orange – hence the description as the '**orange-peel effect**'.
The larger is the value of H (Erichsen number), the larger is the formability of the material's sheet before fracture.
Drawback: the results are variable even with materials of uniform quality; H depends largely upon the applied pressure between the blank-holder and the die-face.
This test is not used anymore.

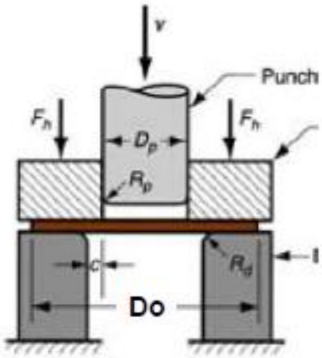


LDR and its relationship with $\overline{R}$

Limiting Drawing Ratio (LDR) = D_0/D_p

where, D_0 is the max diameter of the blank and D_p is the diameter of the punch



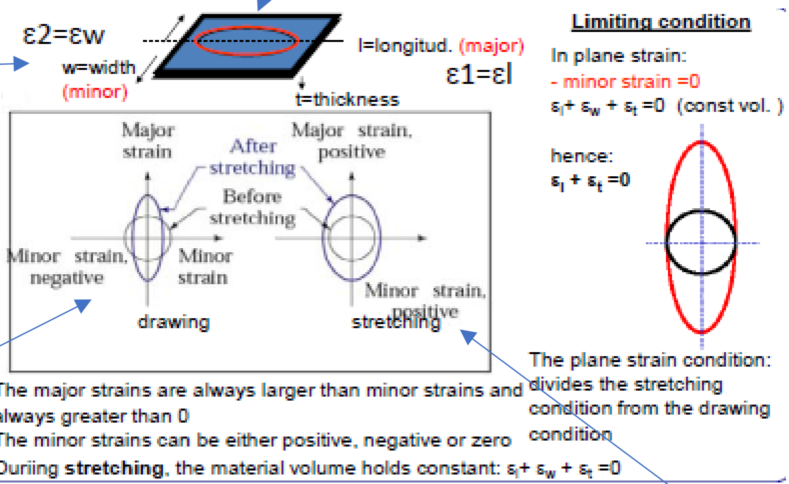


LDR increases as $\overline{R}$ increases → perfectly linear relation between $\overline{R}$ and LDR for various sheet alloys (Max ideal value for LDR = e = 2.718).

Strain states in metallic sheets

Sheet with a red circle drawn on it → after elongation along one direction, it will be an ellipse

Deformation along the longitudinal direction



Elongation

↓

Thinning and necking

↓

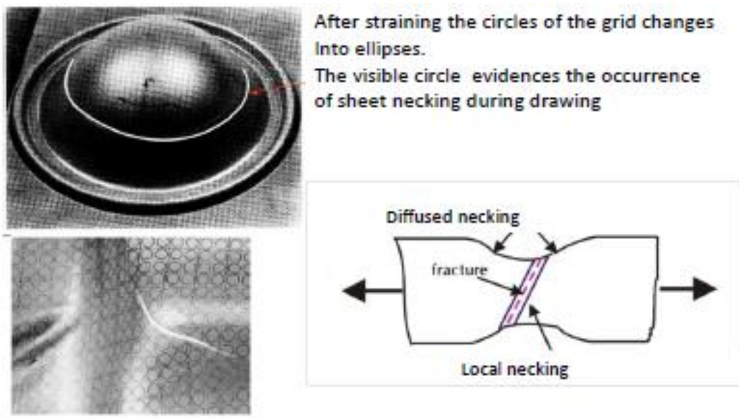
Material may fail

Diameter of the ellipse smaller than the original after the elongation

Diameter of the ellipse larger than the original after the elongation

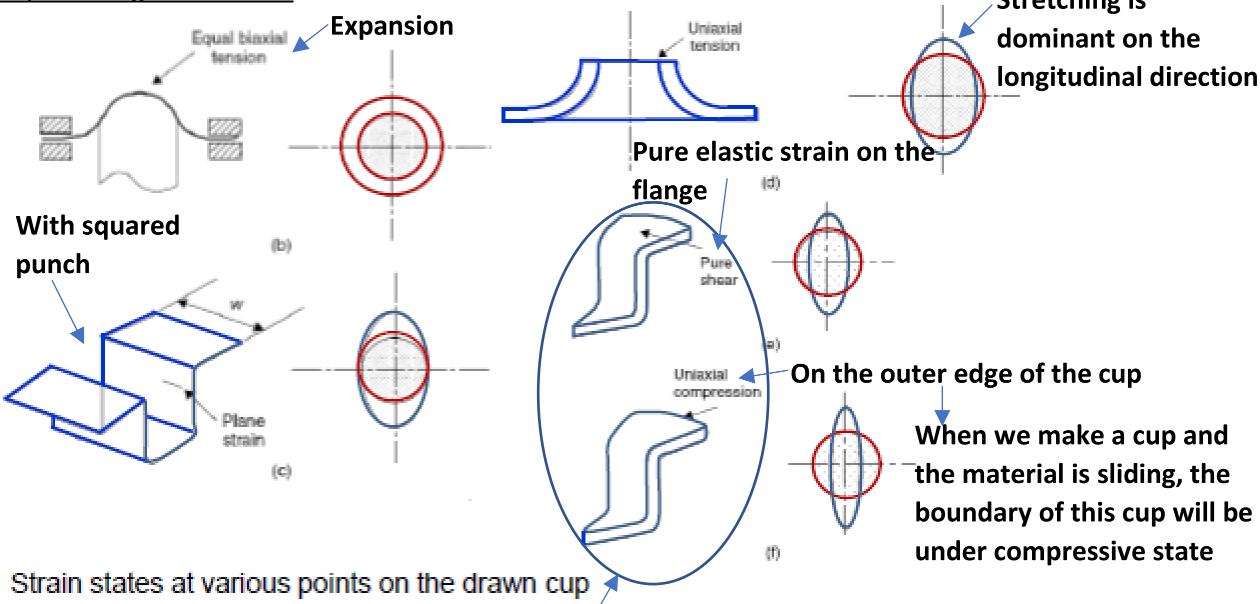
Strain grid and necking detection

Deformation by straining should occur in such a way that necking and thinning are avoided.



Defects → cracks on top of the dome and folding → component must be rejected.
Necking in the sheets along the thickness is like in tensile test, the necking before fracture → necking → we are in triaxial stress state.
The material may have a brittle fracture due to this triaxial stress state.

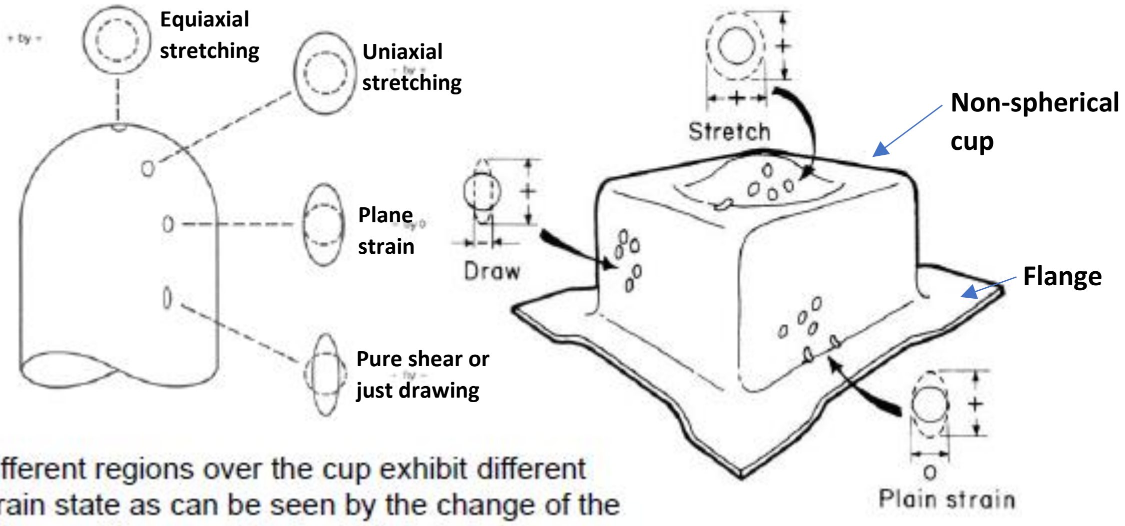
Cup drawing: strain state



Strain states at various points on the drawn cup

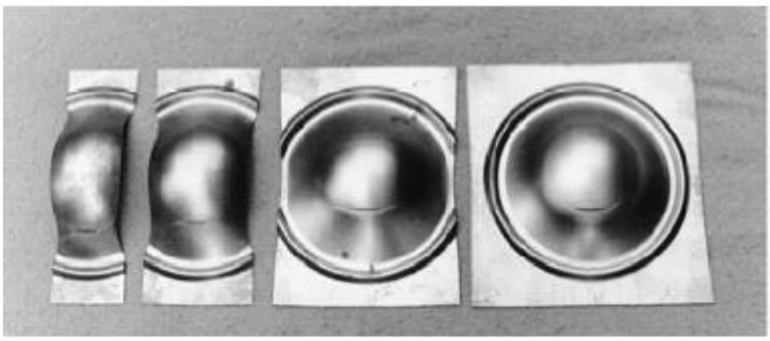
Both are anisotropic drawing but pure shear is less severe than compressive

Cup forming: strain states



Different regions over the cup exhibit different strain state as can be seen by the change of the ellipses with respect to the initial circle

Nakazima test

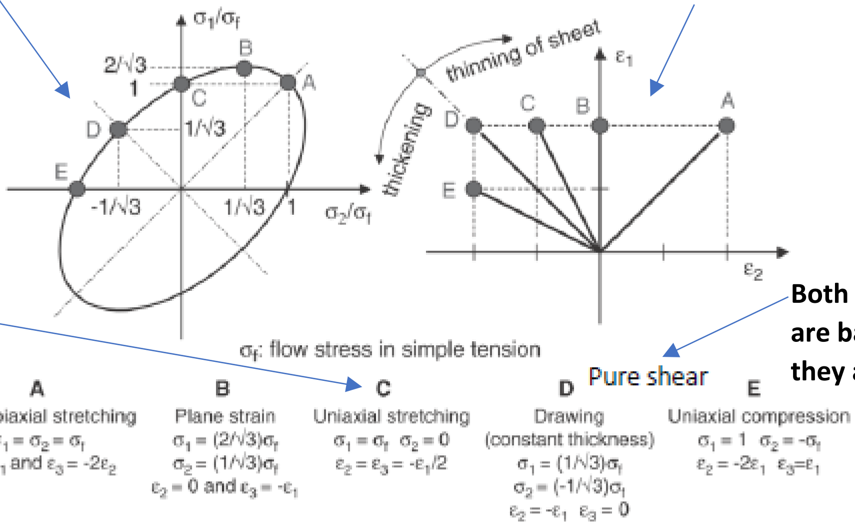


Various strips of different width are cut from an initial circular sheet blank.
The left-most specimen is most likely to be drawn under pure tensile loading. The rightmost specimen is most likely to be biaxially stretched. In place of the die, an oleodynamic pressure-assisted punch is used to shape the specimen (stretching forming).
It is used to obtain a more realistic response of the material formed under actual biaxial sheet drawing process regardless of friction.
After the Nakazima test, a set of cups are obtained; if crack occurs the test fails.

Stress and strain states in sheets

ε_1 and ε_2 recorded after small amount of time to build the diagram

β=ε2/ε1



Tensile because
$\varepsilon_1 = 2\varepsilon_2$

Equipartition of straining (shrinking along the width and the thickness)

$$\varepsilon_w = \varepsilon_t = \frac{1}{2}\varepsilon_l$$

Pure shear

Both strains are balanced, they are equal

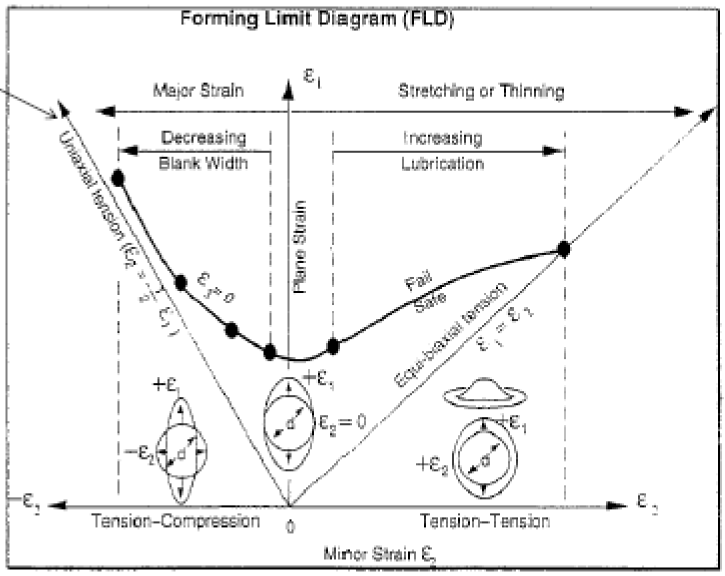
β=1 β=0 β=-1/2 β=-1 β=-2

Stress and strain states of some plane stress conditions (σ3=0) in anisotropic material

Best condition because all the straining goes to the width while the thickness remains constant (no thinning)

Forming Limit Diagram – FLD

R=1

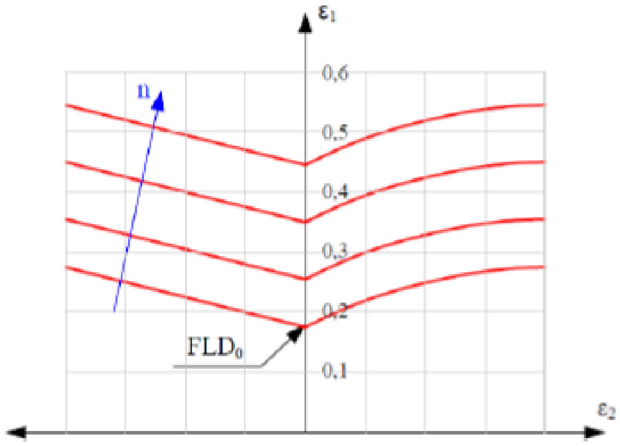


The **FLD** is a test assessing formability of any material especially during deep drawing.
In order to determine whether a given region has failed, a mechanical test is performed by placing a circular mark on the workpiece prior to deformation and then measuring the post-deformation ellipse that is generated from the action on the circle. By repeating the mechanical test to generate a range of stress states, the formability limit diagram can be generated as a line at which failure is onset.

Failure is above the curve because strain larger than the curve will make the sheet fail. Under the curve we have the safe zone → limited → limit for cracking.

Lubrication of the punch (which is flat in deep drawing) gives a better performance with stretching but not it is not needed for drawing (where we have no thickness problem). Frictionless sliding → less impact on thickness during stretching.

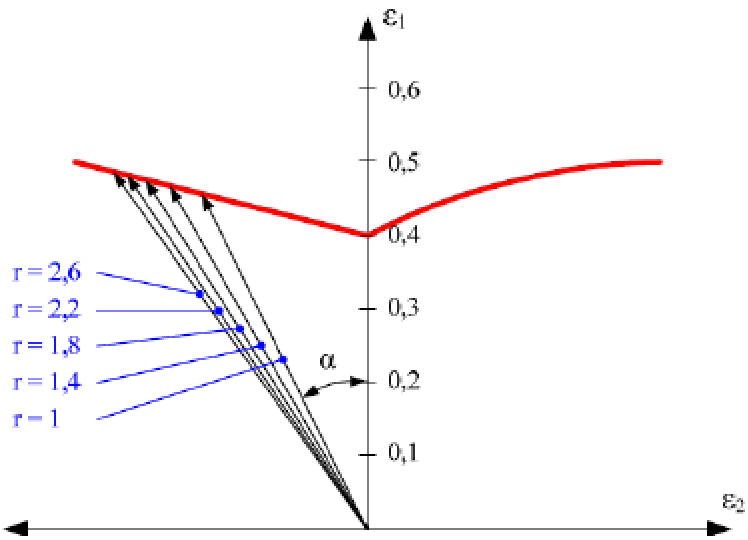
FLD: effect of work hardening coefficient n



$FLD_o = (23,3 + 359 t_o) n / 0,21$ Minimum value of major strain (ε1)

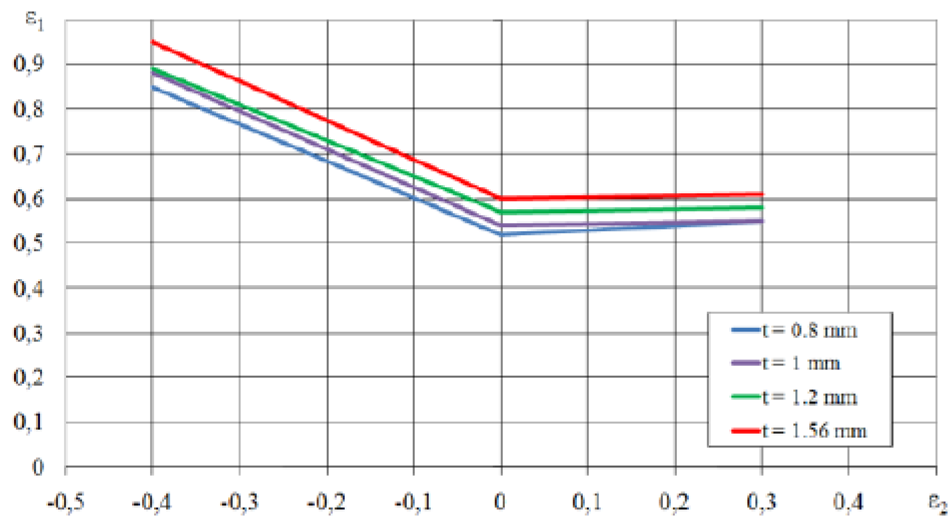
The larger is **n**, the larger is the formability (Oloman law).
We can build a FLD curve for every coefficient n. FLD curves rise with increasing n because both stretching and drawing rise (depending on n) → positive because by rising the curve, the safe zone is extended → extending workability of the sheet in any process of the drawing.
n = uniform strain (plastic) → Larger ductility → Formability improved → It helps drawing and stretching

FLD: effect of normal anisotropy R



R: Resistance to thinning. It doesn't affect stretching but only drawing.
By analyzing the effect of anisotropy coefficient, it was found that higher r values have favourable effect on the limit strain values particularly for ε2 < 0 (we are increasing straining of w2 in negative by pulling).

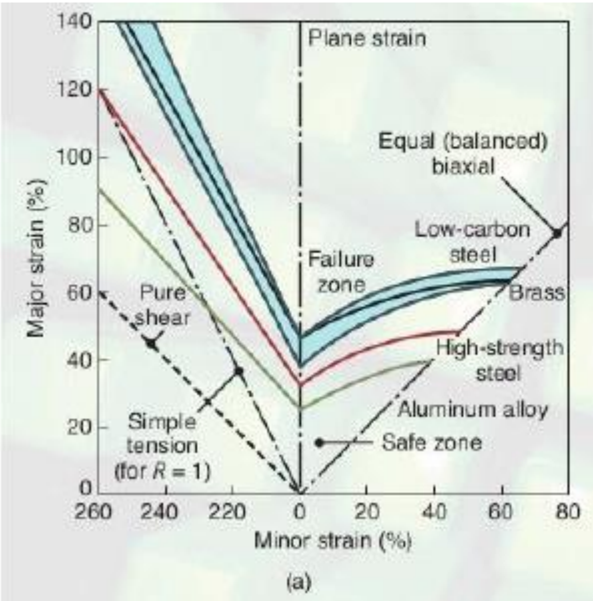
FLD: effect of thickness for DC05



Similar tests were carried out for DC04 and DD14 material grades.

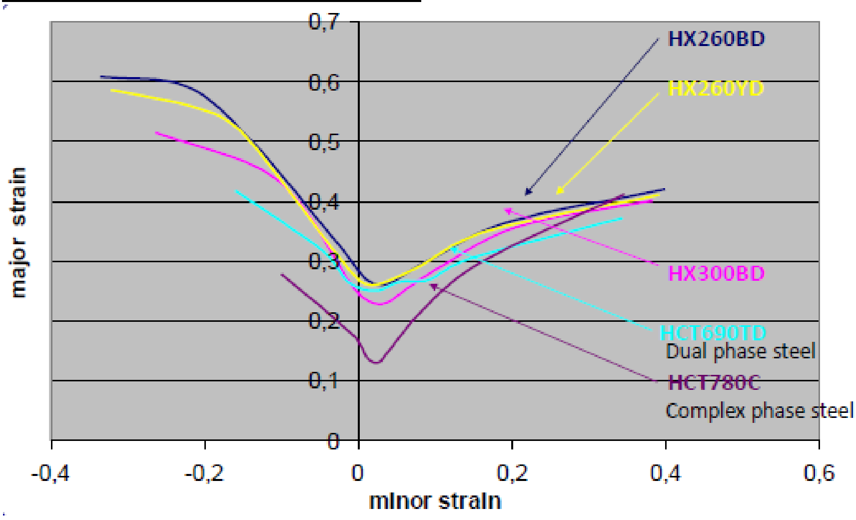
In most of shape forming applications, the thickness is already decided, we can't change it. Changing it means changing the weight, so thickness will be lower and lower, which is not good.
A larger thickness is better because it makes deep drawing favorable and easier.
A lower thickness reduces formability because the thinner the sheet, the higher is the probability of having a crack in both drawing and stretching.

FLD – Flow Limit Diagram



The measurement of the major and minor axes of the deformed ellipses permits to determine the local strain state in the sheet and to locate it on the FLD.
Thus, FLD will show the plastic strain path undergone by the sheet during the actual forming process.
Materials producers provides the FLD curve for each material as data sheet of the material.
Aluminum alloy: one of the worst, because it easily fails during deep drawing, even though it's highly ductile.
Brass behaves within the low carbon range

FLD of Low Carbon and AHSS steels



H → high strength steel multiphase
X → the production process is not known
C → Cold forming
D → Hot forming
T → UTS
No T → yield stress
Higher strength steel → lower formability → it can be shaped but not in complex shapes

Simulated FLD curves: performance of DP steels and BH steels and thicknesses

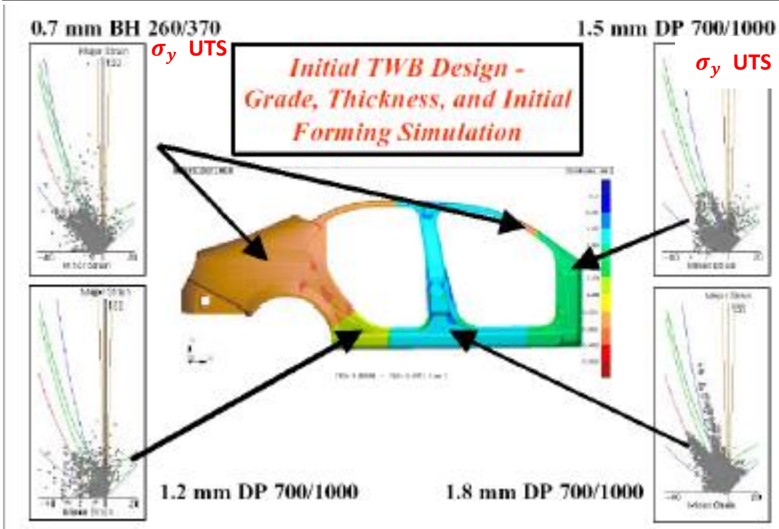


Figure 2. AVC-3-1170 Body Side Outer RH tailored blank layout and initial forming simulation thickness distribution results.

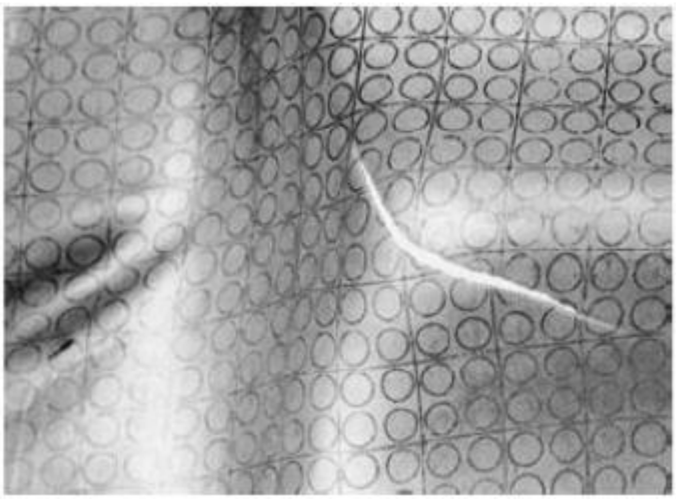
DP: Dual Phase

The part made with tailor welded blanks → different kind of sheets, materials and thickness welded together and shaped into the wanted geometry.

We have to check whether these parts have defects such as necking, cracks. We assess its formability with FLD → dark spots mostly below the curve → safe zone

Deep Drawing: common defects

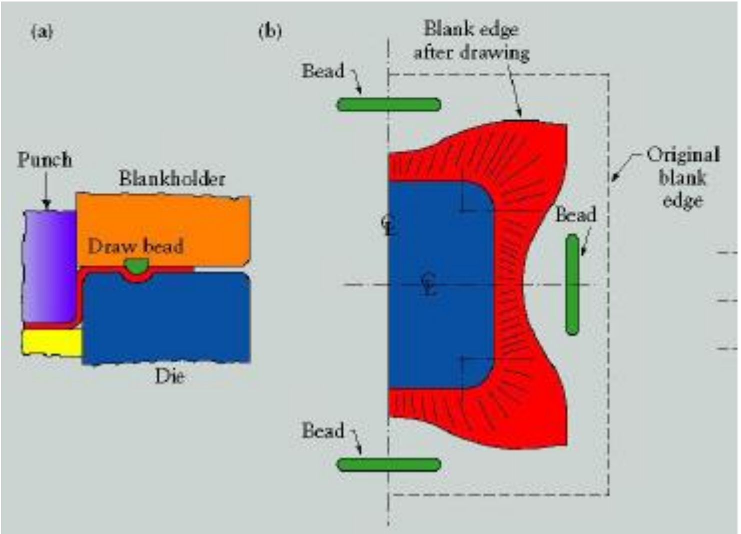
Plastic Instabilities and Fracture in Sheets



STRETCHINGs, FOLDINGs, BUBBLEs are visible, resulting from bi-axial strain state (laterally constrained), with respect to the primary strain direction.

The strain field is sensed by the circle grid,
The grid is impressed over the sheet by selective electrochemical etching.

Drawbeads: to solve wrinkling problems



(a) Scheme of the drawbeads.
(b) wrinkles originated during deep drawing; drawbeads allow to control the flow of the material. *Source*: After S. Keeler.

Drawbeads are inserts put into cavities in order to obtain a local force on the sheet. If we can control the pressure on the blank holder and then on the sheet during drawing, by including the drawbead, we are limiting a sliding of the sheet, because excessive sliding will produce waves → wrinkling

The Wrinkling Limit – WFLD

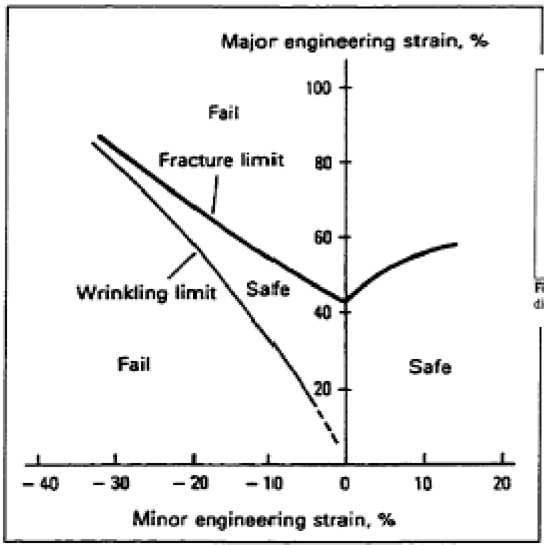


Figure 1.13 Wrinkling limit diagram. (Taylor, 1992)

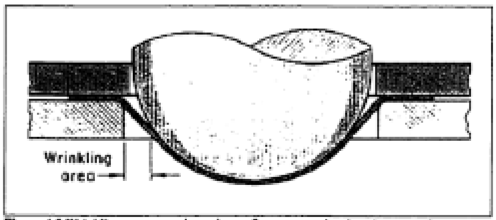
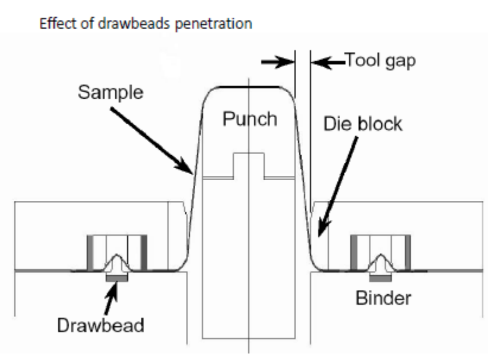
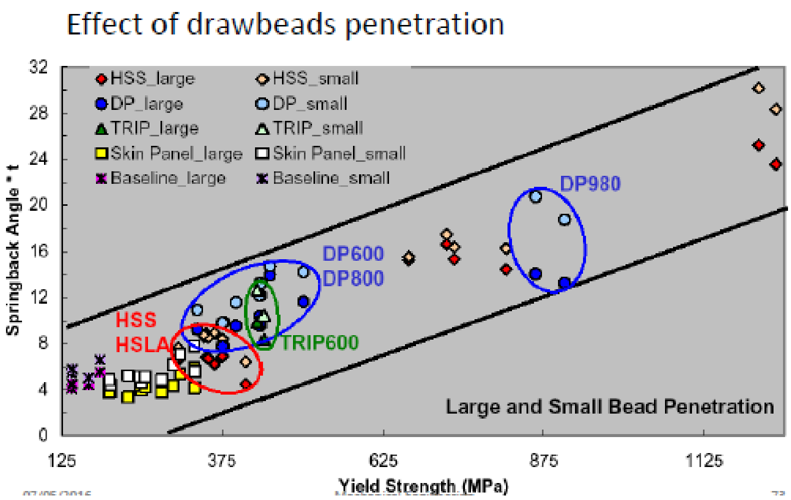


Figure 1.9 Wrinkling can occur in regions of unsupportedregions between the die and punch (ASM, 1976).

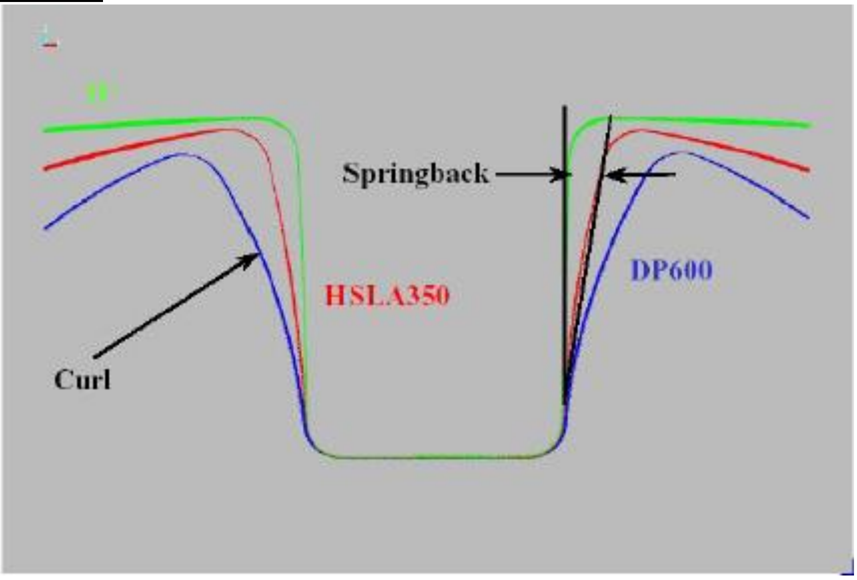
The WFLD curve will restrict the safe part more → higher probability of failure

Springback and drawbeads

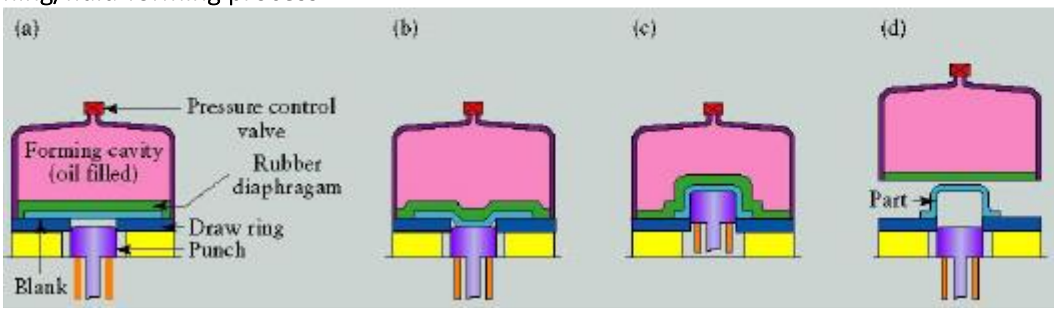




Springback for three steels



Hydroforming/fluid-forming process



Unlike in the ordinary deep-drawing process, the dome pressure forces the cup walls against the punch. The cup travels with the punch, and thus deep drawability is improved.
→ High pressure transmitted to the sheet while its being stopped according to the geometry of the mold

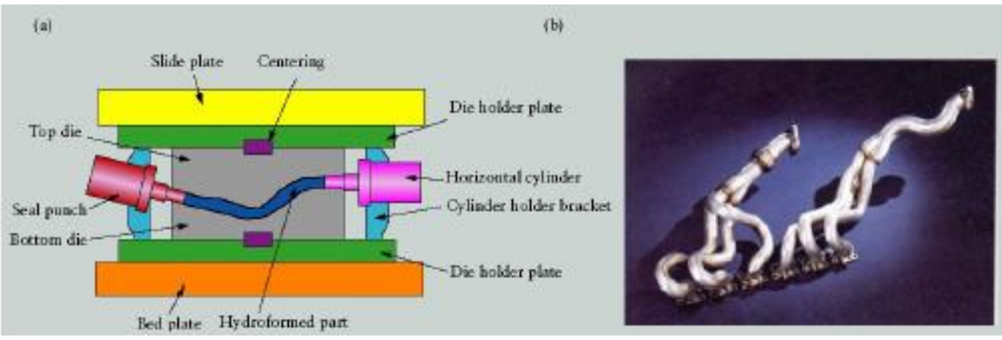
[Hot working is not good for Al panels because at high T the material oxidizes, and we need a solution to drag, to wash, … Hot working has an additional problem: we need to prepare surface for finishing. If we apply cold working instead, the surface is already good, we don't need any further refinement.
To improve formability of Al and other strong alloys by delaying necking and failure hydroforming is used.
Hydroforming:

- **Plastic deformation assisted by a pressure increased slowly** → more redistribution of straining → failure delayed
- **Ultra-fast plastic deformation with high pressure** → heat will remain inside, and material will deform as if it was forming under high temperature. By not providing temperature (energy), the internal friction will produce heat and dislocations will move easily.
- **No friction**]

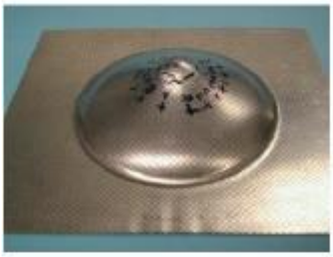
Hydroforming of exhaust gas collector in aluminum alloy

(a) Schematic illustration of the tube-hydroforming process.
(b) Example of tube-hydroformed parts.

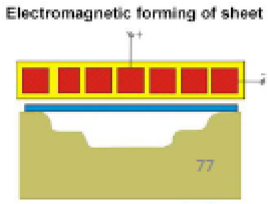
Automotive exhaust and structural components, bicycle frames, and hydraulic and pneumatic fittings are produced through tube hydroforming.
Source: Schuler GmbH.



Electromagnetic forming

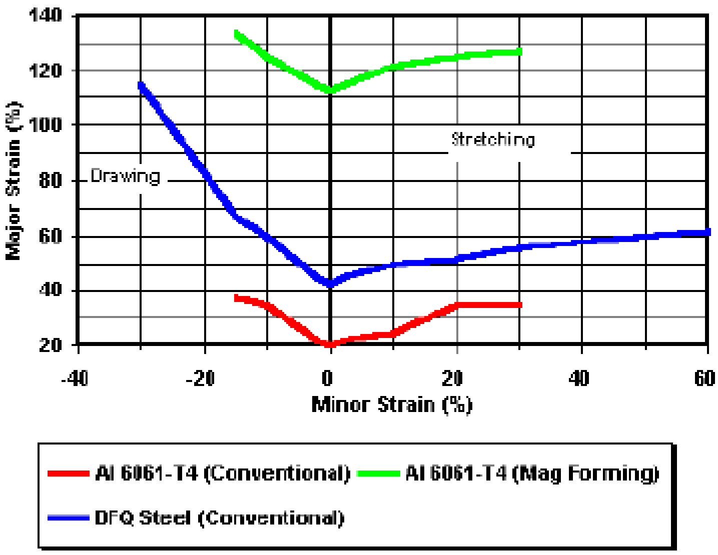


Flat coils for FLD experiments



It exploits fully the mechanism of very fast dislocation movement.
A coil is placed near the metallic piece (electrically conductive metals Al,Cu). The current is flowing into this coil and as a result it generates a magnetic field which accelerates the piece to hyper speed and onto the die → increasing formability without affecting the native strength of the material.

FLD of a special Al-alloy formed by MagnePress electroforming process
To compare performances of alloys undergoing the same process



Typical Strains and Strain-rates for various Processes

PROCESS	True strain	Strain rate m/s
Cold forming		
Forging, Rolling	0.1-0.5	0.1-100
Wire drawing	0.05-0.5	0.1-100
Explosive forming	0.05-0.2	100-100
Hot forming		
Forging, rolling	0.1-0.5	0.1-30
Extrusion	2-5	0.1-1.0
Machining	1-10	0.-100
Deep drawing	0.1-0.5	0.05-2.0
Superplastic forming	0.2-3.0	$10^{-4} – 10^{-2}$

High strain rate → high speed process

High speed but we can deform very small amount of the material

Process done at very low strain rate

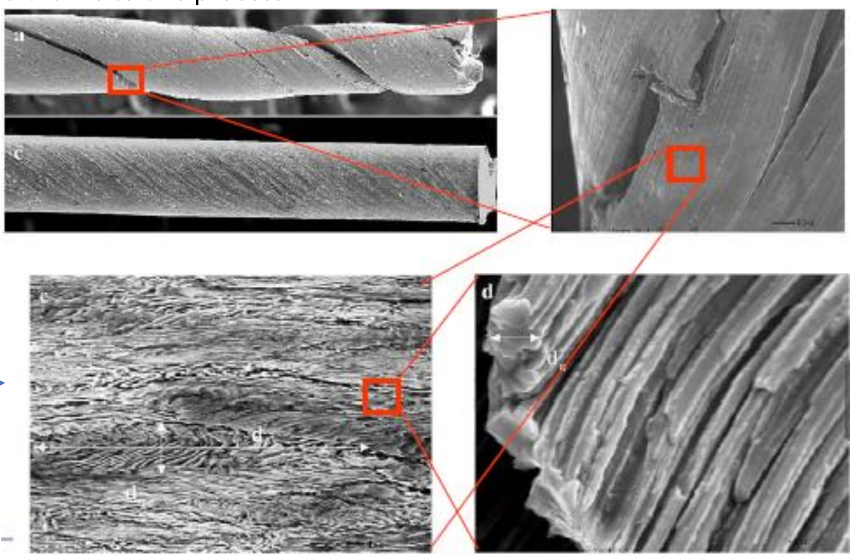
Ti alloy → high strength but also highly formable → superplastic forming at recrystallization temperature where we need very small forces to obtain large strain and to make complicated parts.

Severely drawn eutectoid steel (pearlite) after SEM

Very high strength steels are obtained thanks to this process.

Structures taken from wires drawn by making them pass through a very small calibrated hole

Pearlite → alternating lamellae of ferrite and cementite (Iron and Carbide)



After severe straining, pearlite has been compacted

Superplasticity

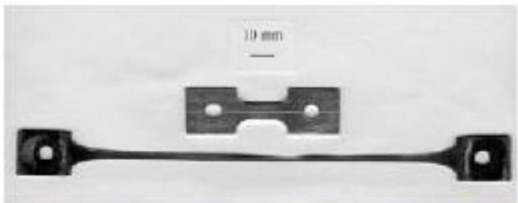
Ability of a material to undergo significant elongation without necking or fracture.

Property related with the capability of the material to plastically deform at high T by Grain Boundary Sliding (GBS) mechanism.

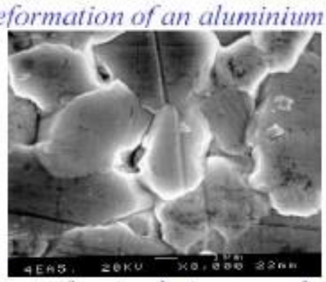
It is industrially used to form complex shaped components or to form difficult-to-form alloys, such as titanium alloys

Peculiar features:

- Very low strain rates (1% per min)
- High temperature (approx recrystallization T)
- Ultrafine grain (UF) (< 10 micron).
- Equiaxed grains, almost spheres



Superplastic deformation of an aluminium alloy



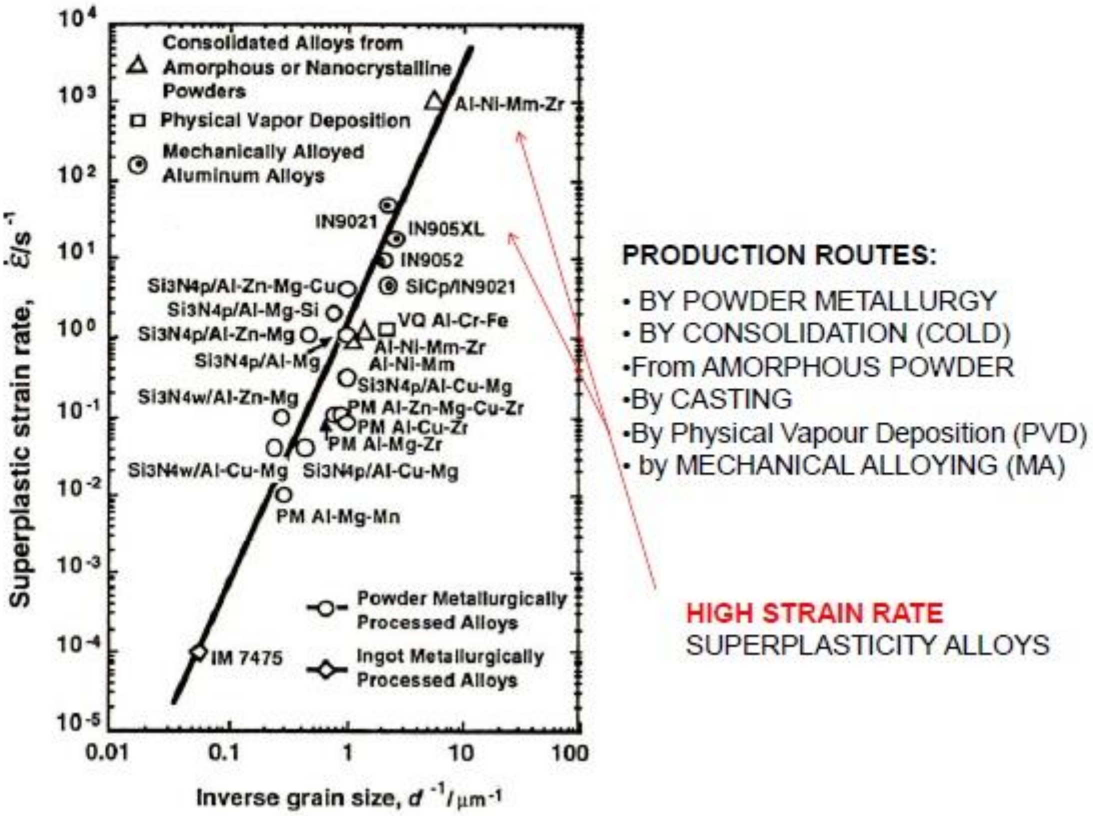
Movement of grains during superplastic deformation of a Pb-Sn alloy

Superplastic alloys: Ti, Al, Inconel, stainless steel.
Assuming a constitutive plastic law of the type:

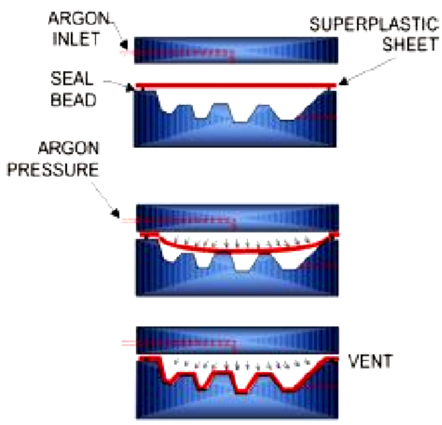
$$\sigma = k\dot{\varepsilon}^m$$

where **m** is the strain rate sensitivity coefficient.
A superplastic behavior is obtained for large m values.
Peculiar features: very low strain rate (1% per min) → because during plastic deformation, the material is straining to same degree and in order to avoid crack we need time for the material to recover, to relax. After the relaxation is completed, we may apply straining. The material will be strained uniformly by delaying crack.
We can make complex geometries.

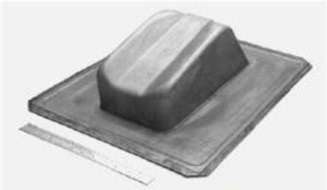
<u>Superplastic alloys</u>



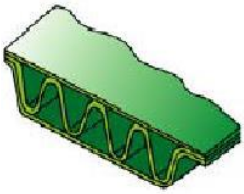
Superplastic Forming (SPF)
SPF is suitable for the production of complex-shaped parts or to form difficult-to-form alloys (e.g. HCP-based alloys, intermetallic alloys, etc.)



Advantages:
• lower tooling costs (at least 80% lower than cold-forming tooling)
• Can be combined with a diffusion bonding to produce complex geometry parts (elimination of mechanical joining: riveting, etc.)
• Reduction of scraps → more rigid parts
• greater process flexibility:
- it allows very small bending radii
- large deep drawing depths
• Elimination of machining processes except for finishing



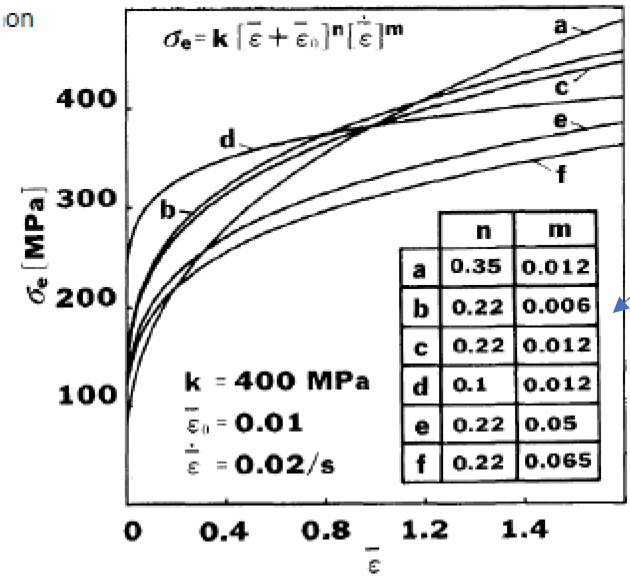
A superplastically formed Al-Li alloy component



SPF employes tooling at approx 800 °C for Ti alloys and a pressurized gas (inert Ar) which acts as a uniformly distributed and strain rate controlled forming force → springback problem doesn't exist at all, we get the net shape.

Strain and strain-rate dependent Constitutive Laws

Extended Ludwik-Hollomon equation:

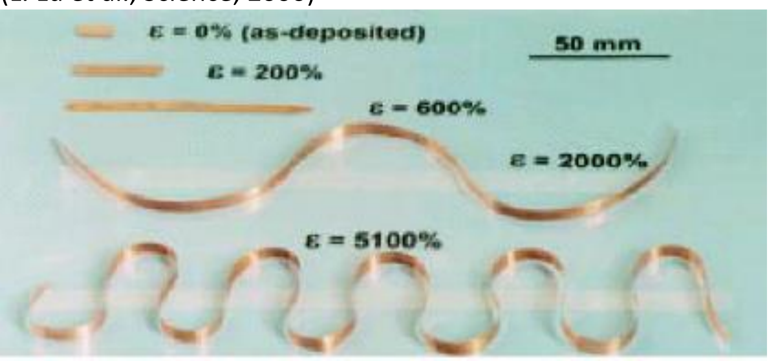


The larger is **m**, the larger is the superplastic behavior.

Different alloys

Superplastic behavior of nCu at room temperature

(L. Lu et al., Science, 2000)



nCu specimen:

- Produced by electrodeposition method
- Shown before and after cold plastic deformation at room temperature under different degrees of deformation.

[Nanometers → small and spherical grains → no possibility for dislocations to move → the plastic deformation that we may observe can be obtained by relative sliding between grains like twinning (rotation of spherical grain allows to have plasticity).

Small grains are accompanied by high energy at gbs → amorphous phase (gbs) become more relevant → material has a very high energy.

→ It is more likely that we are working at high T → rather than having superplastic behavior at recrystallization T, the high energy at gbs makes the alloy operate at room T as if it was at high T → gbs diffusion is huge thanks to the very small size grains.]

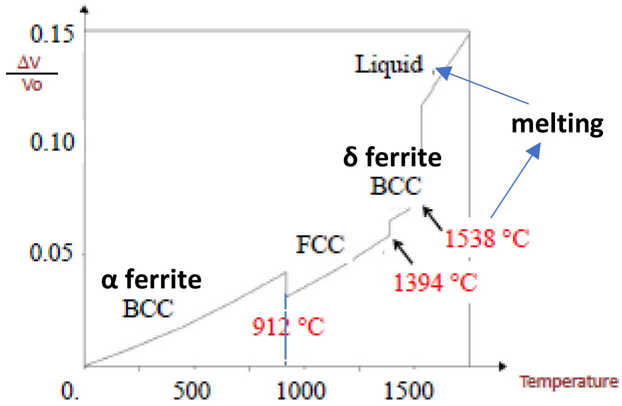
Lecture 9

<u>Fe-Fe$_3$C phase diagram</u>

Fe is important in phase diagram because of its polymorphic behavior.

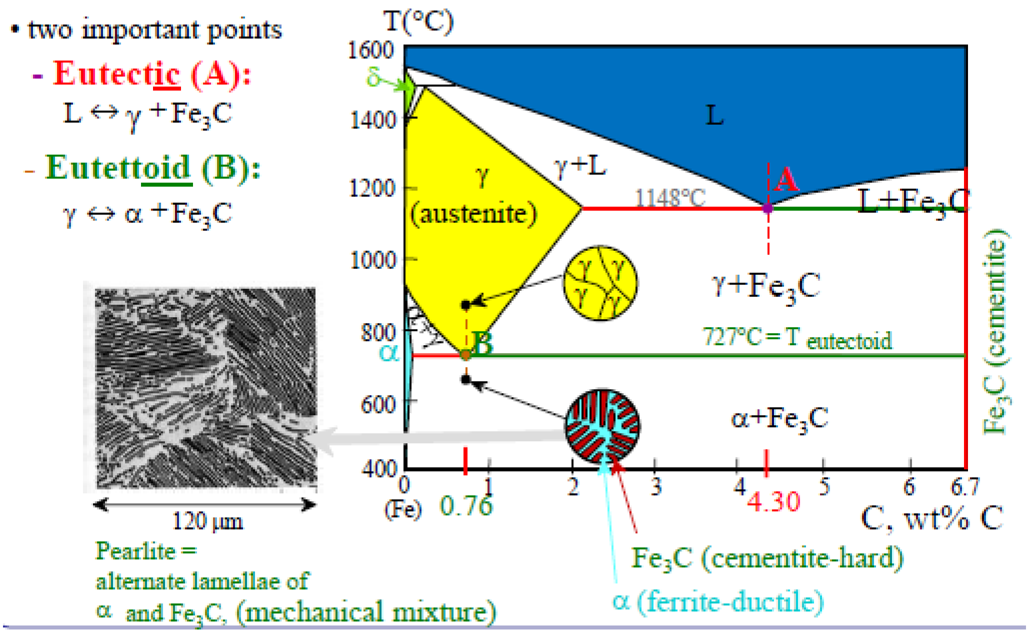
<u>Polymorphic behavior of Fe: density</u>

We have a change in volume as temperature increases → density decreases because of the change from BCC to FCC and then it increases again.

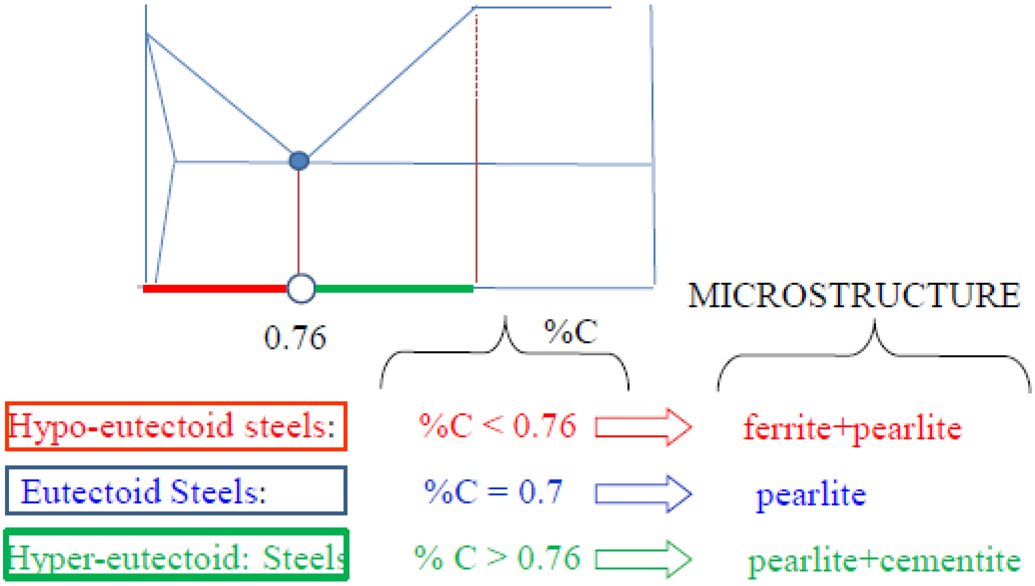


<u>Metastable equilibrium phase Diagram (Fe-Fe$_3$C)</u>

Carbide is not the stable phase of Carbon because it is subjected to change as temperature, … change.

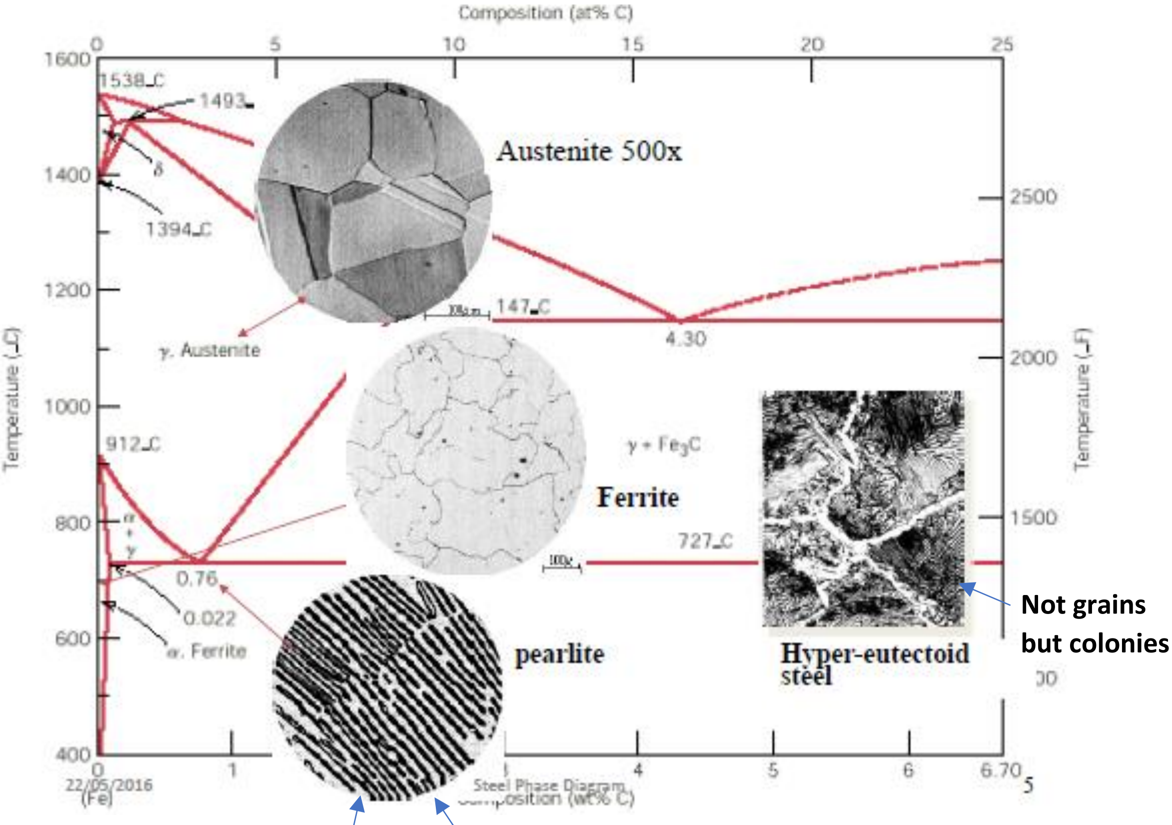


Classification of Steels as a function of %C



Eutectoid steels are also called pearlite steels.

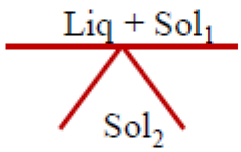
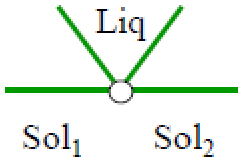
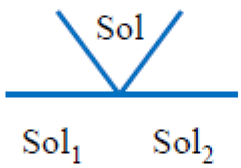
In **hyper-eutectoid steels**, a hard phase (cementite) surrounds a relatively soft phase (pearlite) → detachment of the soft phase → even the soft phase will be brittle in this combination. Therefore, we have strong and hard but not tough steel → not usable. Spheroidizing pearlite and cementite will improve this combination with a more favorable microstructure with well distributed carbides into ferrite.



Lamellar structure

The interlamellar distance is a measure of the fineness of pearlite
→ the smaller the distance, the finer and the harder is pearlite

Invariant points in the Fe-C system

Invariant point	Reaction	phase diagram	location
PERITECTIC	$Liq + Sol_1 \underset{heating}{\overset{cooling}{\rightleftarrows}} Sol_2$		C=0.16%C T_P=1493°C
EUTECTIC	$Liq \underset{heating}{\overset{cooling}{\rightleftarrows}} Sol_1 + Sol_2$		C=4.3% T_{EU}=1147°C
EUTECTOID	$Sol \underset{heating}{\overset{cooling}{\rightleftarrows}} Sol_1 + Sol_2$		C=0.76% T_{ED}=727°C

Fusion welding → most of the structure steels used for construction must melt at peritectic temperature because they have low %C.

Equilibrium Critical Transformation Points

- A_1 → pearlite (FCC)/austenite (BCC) transformation
  C = 0.76%, eutectoid temperature (727°)
- A_3 → austenite/ferrite transformation curve
  C < 0.76%, A3 line (910°)
- A_{cm} → austenite/cementite transformation line or solubility limit line of C precipitating as secondary carbide phase
  C > 0.76%

Ferrite solubility limit line: solubility line of C in ferrite precipitating as tertiary carbide phase

Non-Equilibrium Critical Transformation Points

At high heating and cooling rates, we can't follow the phase diagram anymore and the temperatures will be A_c (chauffage) higher than A and A_r (refroidissment) lower than A. As the rate increases, we have a delay of response.

$A_{r1}, (A_{r3})$ Temp. Of end (initial) formation of ferrite from austenite upon fast cooling

$A_{c1}, (A_{c3})$ Temp. of initial (end) formation of austenite from ferrite upon fast heating

$B_s, (B_f)$ Temp. of d'initial (end) formation of bainite, from austenite, during cooling (moderately fast cooling) …

$M_s, (M_f)$ Temp. of initial (end) formation of martensite from austenite upon very fast cooling

$A_{r,cm}$ Temp. of initial formation of cementite from austenite opon slow cooling

$A_{c,cm}$ Temp. of initial formation of austenite from cementite upon slow heating

Influence of alloying elements on transformation points

Austenitizing/Ferritizing → some alloy elements, when added into the solid solution, may promote the expansion of the field of either austenite or ferrite. Austenitizing also means that if we are going to cool any steels from austenite condition and this cooling is not very severe, these elements may help austenite to stabilize at room temperature.

Strong AUSTENITIZING: Ni, Mn, Co (at room T)
Weak AUSTENITIZING: C, N, Cu
Cu improves the corrosion resistance but, if used in larger amounts, it may induce brittleness.
Strong FERRITIZING: Cr, Si, Al, Mo, Ti, V
Weak FERRITIZING: B, S, Zr, Nb, Ce, Ta
Equation to estimate critical point A_{c1} after we add such alloy elements:

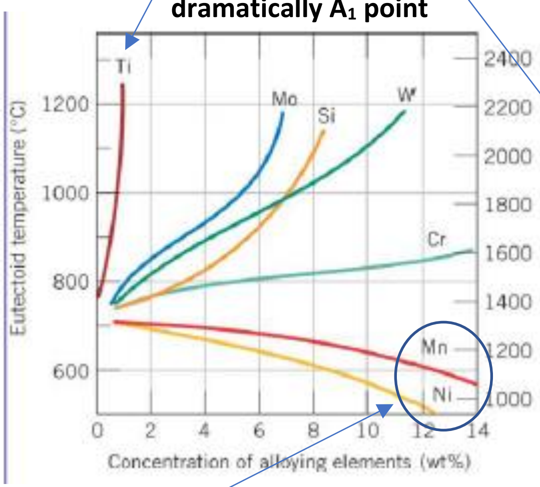
$$A_{c1} = 723 + 16.9*(1.72\ Si - 0.63\ Mn + Cr - Ni + 0.37\ W + 17.2\ As)$$

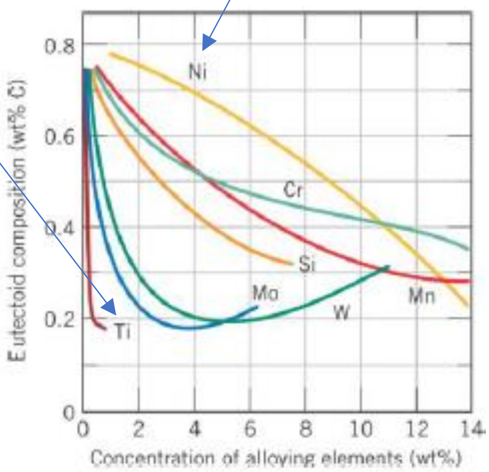
$$A_{c3} = 910 - 203(C^{0.5} + 0.15\ Mn + 0.05\ Cr + 0.075\ Ni + 0.1\ Cu - 3.45\ P - 0.22\ Si - 0.16\ Mo - 0.5\ V - 0.06\ W - 0.58\ As - 1.97\ (Al+Ti)$$

Influence of alloying elements on A_1 and C_E

%C decreases as Ni amount increases

A small amount of Ti affects dramatically A_1 point

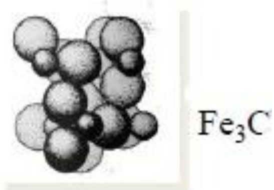


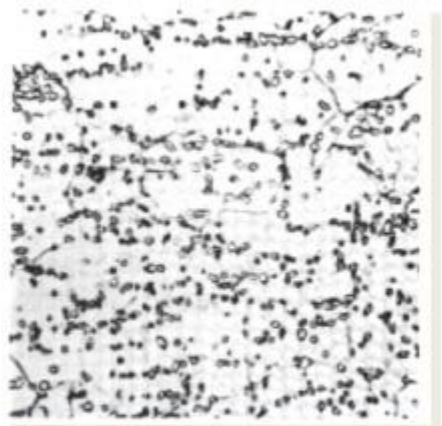


Austenitizing elements decrease A_1 temperature as their amount increases.
Changing A_1 (temperature at which austenite changes to pearlite) → these elements may affect the phase transformation in cooling.

Cementite

- **Very hard compound** (VHN=1200)
- **Brittle** (plastic elong. = 0)
- Fe_3C: **orthorhombic Structure**, with **4 at. of C and 12 of Fe**.
- Contains **6.67 w% C** (25% at.)
- It is present in all steels when C > 0.025% in different morphologies such as (pearlite, bainite) **lamellae**, **needle** (martensite), **spheroids** (after H.T.), or as **film** around austenite grain boundaries (in **Hyper-eutetoidic steels**).
- It markedly affects mechanical strength of steels (it increases the strength).
- In alloyed steels the iron cementite may be replaced by complex carbides such that Fe → **(Cr, Mo, Mn).**





Fine spheroidal carbides in ferritic matrix in hyper-eutectoid steels (500x)

Graphite

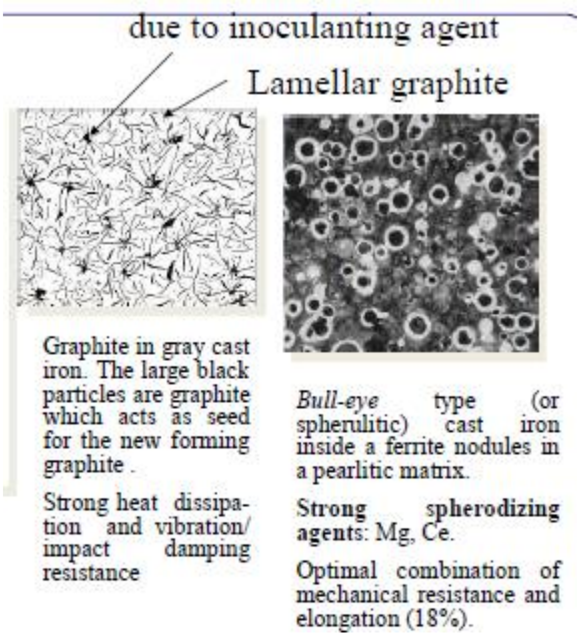
Graphite has **lower density** than cementite and it is not a metallic phase.

It is promoted by:

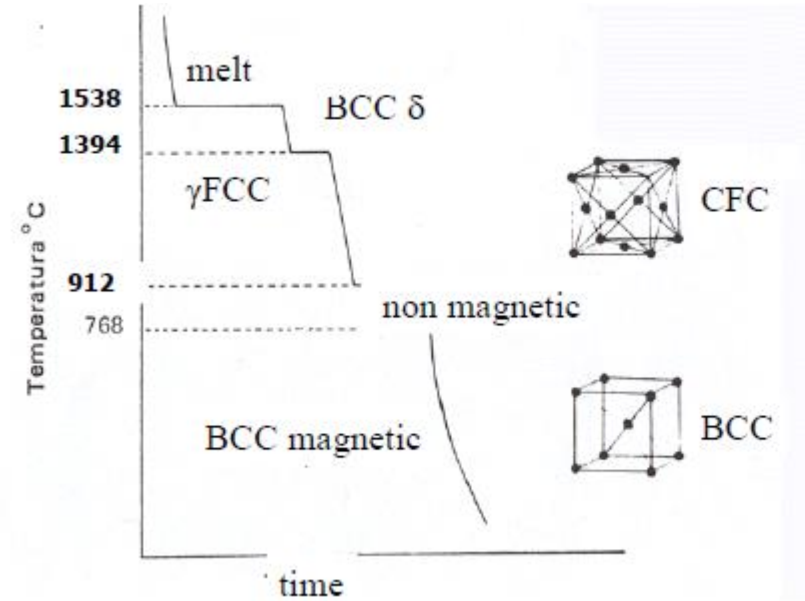
- **extremely slow cooling**
- **high concentration of C** (>2%)
- **+ Si** (1-3 %) (as in cast iron).

In oxidizing environment, graphite has very low friction coefficient, which acts as lubricant in machining operations (high oxidization resistance and high corrosion resistance in the case of cast iron).

Friction coefficient increases with increasing temperature under both vacuum and controlled atmosphere.

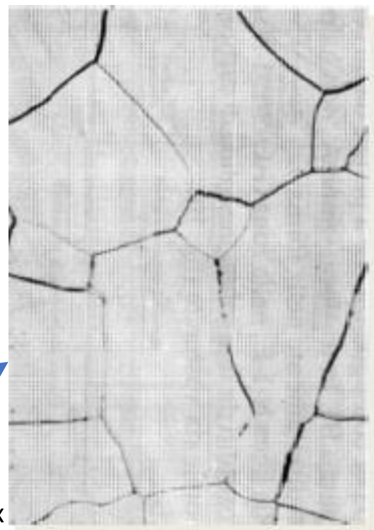


Allotropy of Fe (Transition of Fe)



Ferrite

- Fe phase
- BCC structure.
- The most stable at room temperature;
- it is often obtained on cooling transformation from austenite
- C has very low solubility in this phase; the lowest is at room temperature (**C ≤ 0.0025%**).
- Fe atoms can be replaced in the crystal lattice by other alloying elements – e.g. **Cr, Mo, Si**.

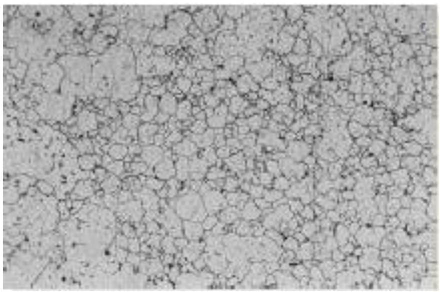


Grains are quite rounded polycrystals

Ferrite 500x

Austenite

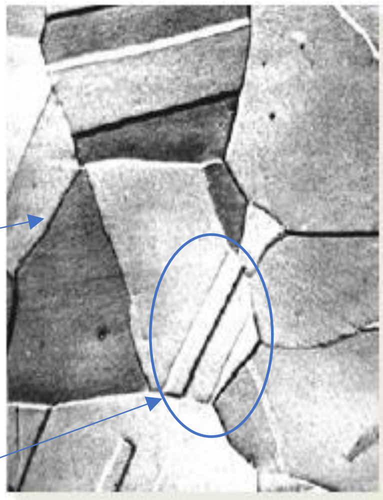
On heating at high temperature, the BCC lattice changes to FCC (austenite). When austenite is formed the solubility of any interstitial (e.g. C) into austenite increases Fe of austenite can be replaced by **Ni** and **Mn**, which easily adapt to the crystal lattice of Fe without significant lattice distortion.



Grains are sharper than ferrite

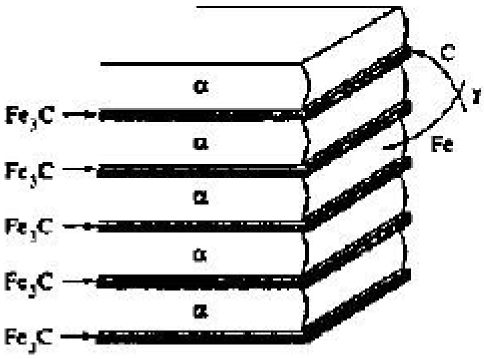
Austenite is stable at room T by adding a large amount of alloy elements.

Twins



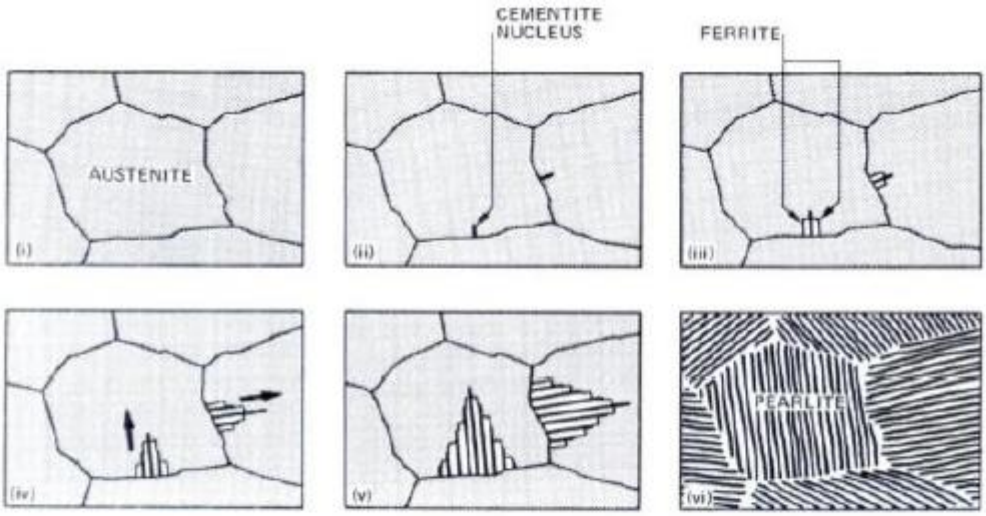
Growth of pearlite (eutectoid) in steels

Pearlite is not a phase like ferrite and austenite but a mixture of two phase, $\alpha + Fe_3C$ (alternating lamellae).

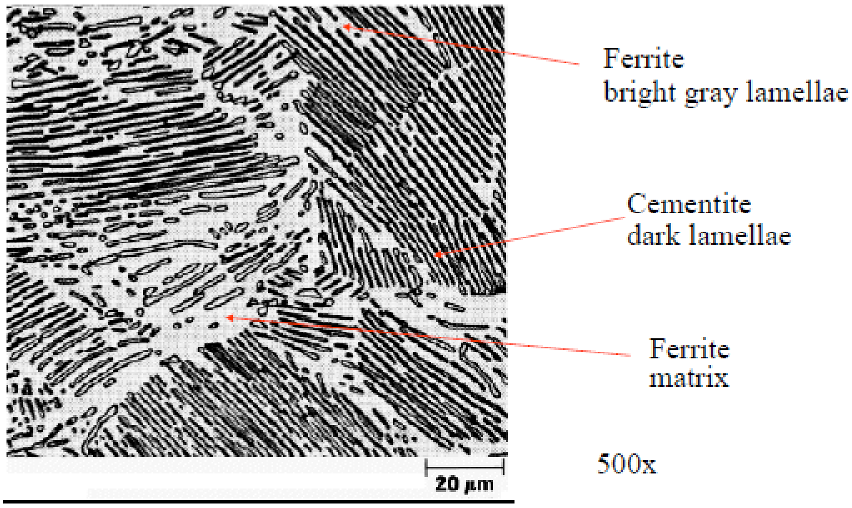


On cooling, when austenite reaches A1 temperature, the austenite totally transforms into pearlite with a similar mechanism of eutectic systems.

Nucleation and Growth of pearlite

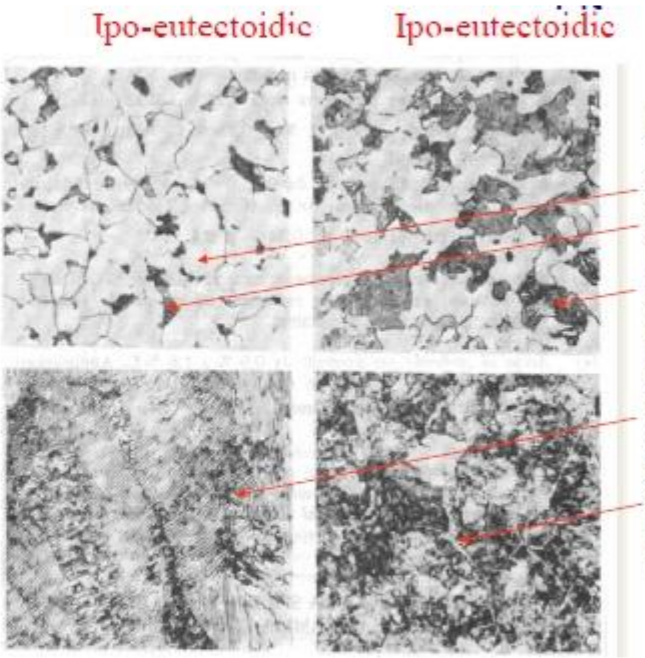


Pearlite: microstructure



Pearlite morphology using SEM

Plain Carbon steels



Micrographs of plain carbon steels.

a) Low carbon steel with 0.16 % C: ferrite (gray) with small amount of pearlite (dark) (500x);

b) iper-eutecoid steel with 0.35% C: with increasing amount of pearlite colonies (500x);

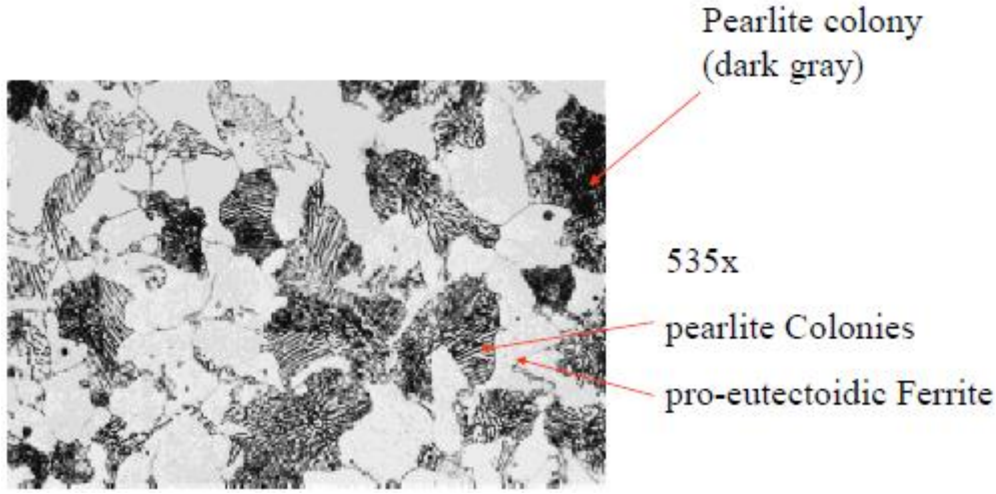
c) Near-eutectoid steel 0.85 %C: showing coarse pearlite lamellae (1000x);

d) Iper-eutectoid steel with 1.3 %C: it shows thin intergranular film of cementite (light gray) between pearlite colonies (500x)

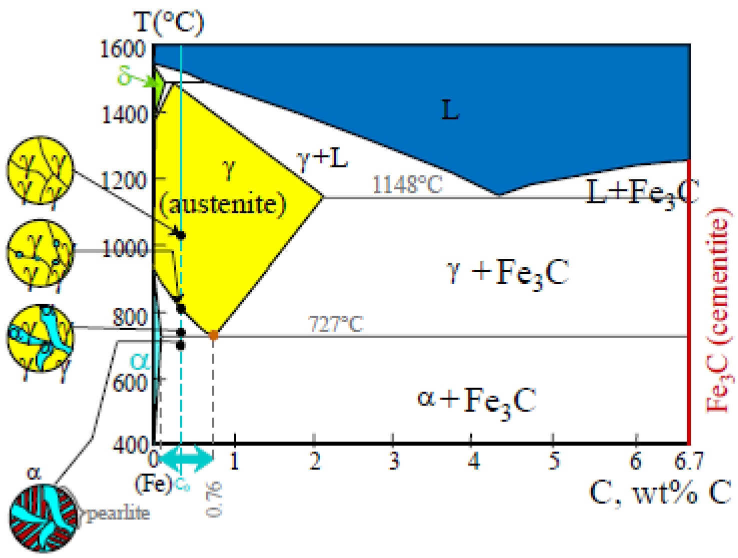
Larger amount of C converts into pearlite → amount of pearlite increases

At eutectoid point, all austenite disappears in favor of pearlite.

Hypo-eutectoidic Steel: 0.38%C



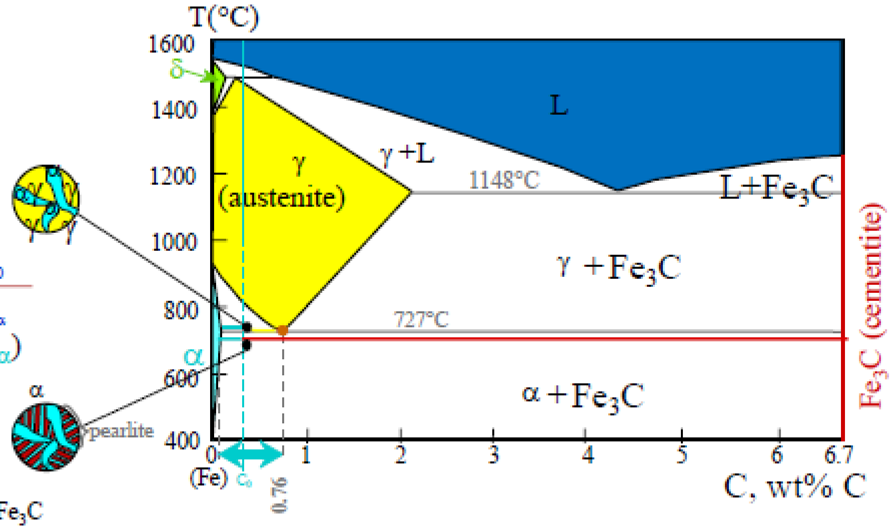
Transformation on cooling: Hypo-eutectoid Steel



Lever rule on cooling: Hypo-eutectoid Steel (C < 0.76%)

$$W_\alpha = \frac{C_\gamma - C_0}{C_\gamma - C_\alpha}$$

$$W_\gamma = (1 - W_\alpha)$$

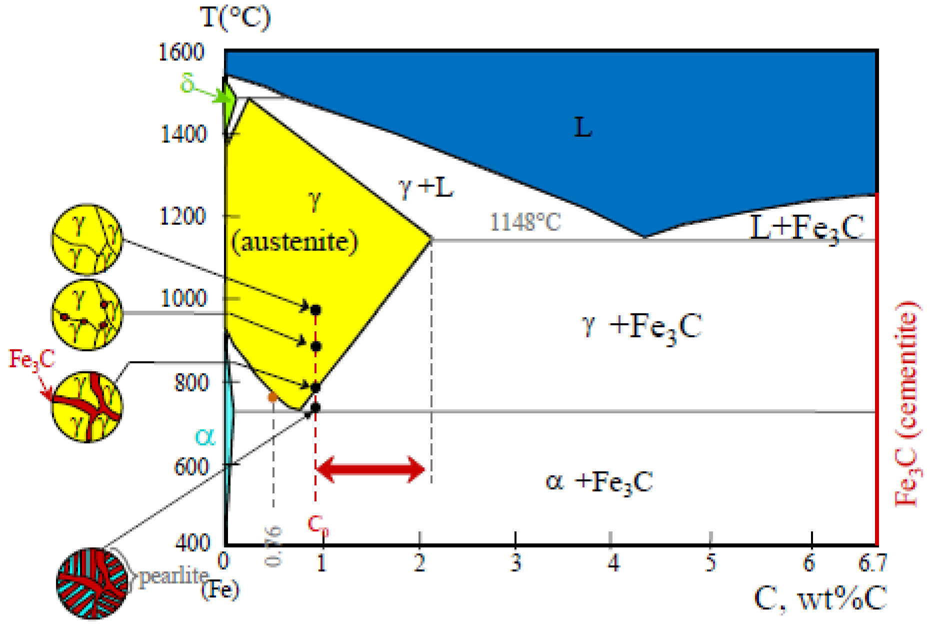


pearlite $= \alpha + Fe_3C$

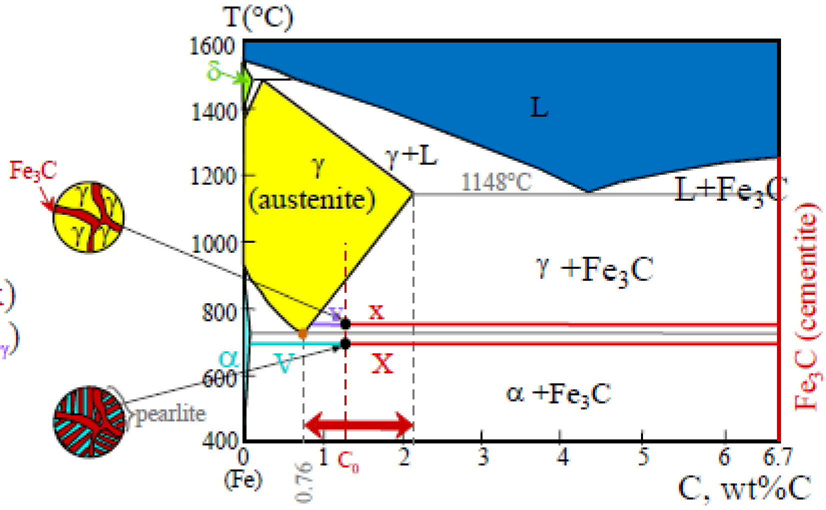
$$W_{\alpha'} = \frac{C_{Fe3C} - C_0}{C_{Fe3C} - C_\alpha}$$

$$W_{pear} = (1 - W_{\alpha'})$$

Transformations on cooling: Hyper-eutectoid Steels (0.76% < C < 2.1%)



Lever rule on cooling: Hyper-eutectoid Steels (0.76% < C < 2.1%)



$W_\gamma = x/(v + x)$

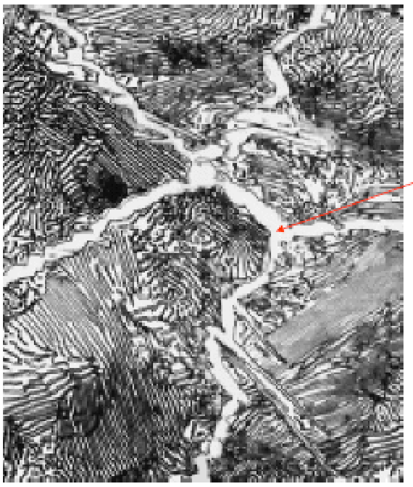
$W_{Fe_3C} = (1 - W_\gamma)$

$W_{pearlite} = W_\gamma$

$W_\alpha = X/(V + X)$

$W_{Fe_3C'} = (1 - W_\alpha)$

Cementite in Hyper-eutectoid Steels



Pro-eutectoid cementite which envelops pearlite colonies

1000x